Business Cycles in the Data Economy*

Carl-Christian Groh† Oliver Pfäuti‡ Farzad Saidi§

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Abstract

How does firms' use of data change the propagation of business cycles? Using employer-employee data to measure firm-level data intensity, we document that investment cyclicality is U-shaped in data intensity: firms with no data-related employment and highly data-intensive firms respond more strongly to business cycle shocks than firms in between. We rationalize this pattern in a tractable model in which firms accumulate data directly or as a by-product of production via a data feedback loop. Data raises expected productivity, dampening firms' relative exposure to aggregate shocks, and lowers uncertainty and hence investment costs, making data and capital complements. This complementarity strengthens with data access and overturns the dampening at the top of the distribution. Embedded in a New Keynesian general equilibrium model, it amplifies the transmission of aggregate shocks, with the data feedback loop as the dominant channel. Mapping digital markets regulations into the model, we evaluate their cyclical effects.

Keywords: data, uncertainty, investment, business cycles
JEL codes: D21, D81, E22, E32

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†Department of Economics, University of Bonn, carlchristian.groh@gmail.com
‡Department of Economics, University of Texas at Austin, pfaeuti.oliver@gmail.com
§Department of Economics, University of Bonn, saidi@uni-bonn.de

1 Introduction

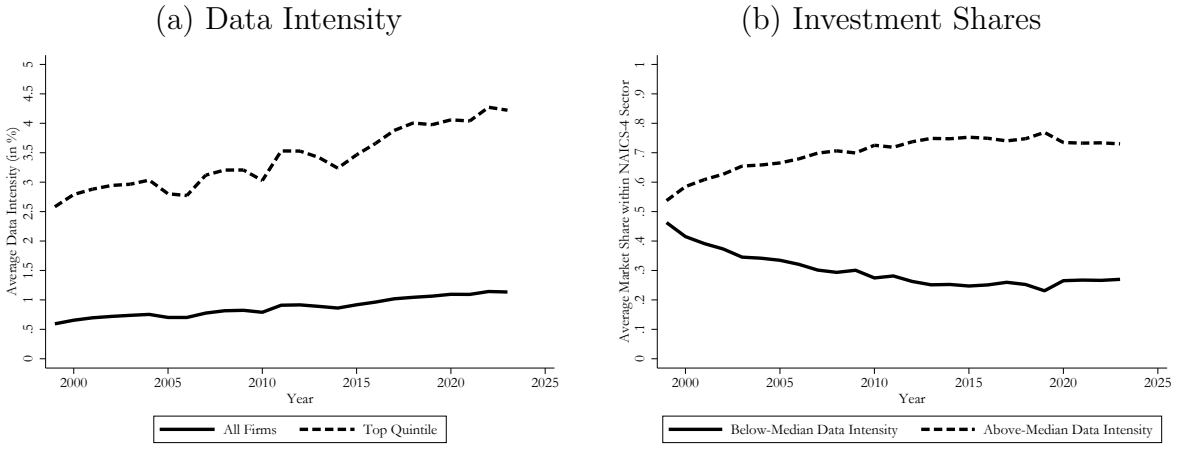
As firms increasingly rely on big data to forecast demand, productivity, and other payoff-relevant variables, data use is inevitably going to change the propagation of business cycle fluctuations. By 2015, roughly three-quarters of U.S. manufacturing firms were already implementing some form of predictive analytics (Brynjolfsson and McElheran, 2019). The economic value of data has grown rapidly over the last few years (Abis and Veldkamp, 2024), motivating a growing body of work on the macroeconomic implications of data (e.g., Farboodi and Veldkamp, forthcoming). The rapid advances in generative AI further increase the importance of big data by enhancing the value of data for predictive analytics (Agrawal et al., 2019). Yet it remains largely unexplored how firms' ever-increasing reliance on data reshapes the transmission of aggregate shocks to their investment.

This paper studies how heterogeneity in firms' data intensity affects the transmission of business cycle fluctuations to investment. To do so, we exploit a novel measure of firms' data utilization based on the occupational composition of their workforce. Figure 1 documents the growing importance of data-intensive firms for aggregate investment. Panel (a) shows that firms' data intensity has risen sharply over the past two decades: the average share of data-related employees has doubled, and it has risen by about two percentage points among firms in the top quintile. Panel (b) shows that this has coincided with a substantial reallocation of investment shares: whereas firms above and below the within-year median data intensity accounted for similar investment shares within industries in 2000, data-intensive firms account for nearly three-quarters of industry investment by 2023.

To measure firms' data utilization, we use employment biography data encompassing the universe of public LinkedIn profiles. This linked employer-employee dataset includes a task-based classification of job roles, which allows us to identify data-related jobs and compute U.S. firms' *data intensity*, i.e., the share of data-related employees in total employment (on LinkedIn). We link this to Compustat data on balance sheets and capital investment, and identify business cycle shocks using the "main business cycle shock" from Angeletos et al. (2020). Using our firm-level information and the aggregate shock series, we then estimate how firms' lagged data intensity shapes the transmission of these shocks to firm investment.

Our main empirical finding is that the relationship between firms' data intensity and their investment responses to business cycle shocks is U-shaped. Investment cyclicality first declines with data intensity, then reverses and rises significantly at the top of the data-intensity distribution. The pattern is robust to controlling for leverage, age, cash-to-assets ratios, intangible-asset ratios, size, and their interactions with the business cycle shock, ruling out that our estimates capture other known transmission channels (e.g., Ottonello and Winberry, 2020; Cloyne et al., 2023; De Ridder, 2024). As more firms move into the

Figure 1: Investment Shares of Data-Intensive Firms



Panel (a) shows the average level of data intensity over time, where data intensity refers to the sample-weighted share of data-related employees (with job roles classified as "data analyst," "data engineer," "data scientist," or "database administrator") at the end of the previous year, separately for all firms and for firms in the top quintile of the respective distribution in a given year. Panel (b) shows the average investment market share, in terms of capital expenditures, within a four-digit NAICS industry over time, separately for firms with above-median and below-median values of data intensity in a given year.

upward-sloping region of this relationship, further data deepening is likely to raise aggregate investment cyclicality.

To understand and rationalize these results, we develop a model of firm investment with endogenous data acquisition. Data raises a firm's expected productivity and lowers the variance of its productivity—both empirically well-documented facts (Bajari et al., 2019; Corrado et al., 2022; Wu, 2023). A firm's cost of capital rises with its productivity uncertainty, consistent with what we find empirically. Firms can directly acquire data at a cost, and in addition, firms generate data as a by-product of production through a data feedback loop (as in Farboodi and Veldkamp, forthcoming).

We show that our model can rationalize the U-shape in the data. There are two countervailing effects of higher data intensity on investment cyclicality. On the one hand, more data reduces a firm's relative exposure to aggregate productivity shocks because more data increases the firm's idiosyncratic productivity and, hence, makes the firm less dependent on aggregate productivity. On the other hand, capital and data are complements. Aggregate shocks that raise investment also increase the value of acquiring data, which feeds back into firms' investment decisions. This complementarity is particularly important for firms with more data, because these firms are larger and, thus, adjust their capital stock more strongly in response to the shock, thereby leading to a relatively large change in the amount of data these firms acquire. This endogenous adjustment of data can overturn the lower cyclicality associated with better predetermined data access and generate the strong investment

cyclicality of highly data-intensive firms.

To examine the aggregate consequences of these findings, we then embed the data-investment block into an otherwise standard New Keynesian model with capital. Data-using producers accumulate capital and data, acquiring data both directly and as a by-product of production through the data feedback loop. Data materially amplifies the transmission of aggregate shocks: in response to a positive productivity shock, output and investment rise substantially more in the baseline economy than in an otherwise identical economy without data. We calibrate the model such that the data feedback loop and active data acquisition both account for half of the steady state data stock. However, the two channels differ in their importance for the transmission of shocks. The data feedback loop is the dominant amplification channel: shutting it down brings the responses close to the no-data benchmark, far more so than eliminating direct acquisition alone. The reason is that the data feedback loop ties the data stock endogenously to capital, which itself expands in response to the shock. Furthermore, the two channels through which firms can accumulate data are complements: the total amplification of aggregate shocks in the data economy is substantially larger than when considering the two channels in isolation.

Our analysis also allows us to assess the implications of digital markets regulation for business cycle fluctuations—a dimension the policy debate has largely ignored to date. Our empirical findings indicate that mandates requiring dominant firms to share data with competitors have countervailing effects: improving the data access of data-poor firms weakens their responsiveness to business cycle shocks, whereas improving the data access of already data-intensive firms can amplify it. Data sharing measures therefore amplify aggregate fluctuations if they primarily benefit firms with intermediate access to data, and dampen them if they primarily benefit firms with little or none. Our general equilibrium results further suggest that measures restricting the direct acquisition of data, such as the GDPR's data-minimization principle, dampen cyclical fluctuations by less than measures weakening the data feedback loop, such as provisions limiting the internal reuse of data across business lines.[1]

Related literature. Our paper is related to five strands of the literature. First, our work contributes to the growing literature on the relevance of digitization and data for macroeconomic outcomes. Veldkamp and Chung (2024) provide an overview of the role of data in the economy. Empirical studies such as those by Brynjolfsson et al. (2023) and Babina et al. (2024) document that the utilization of big data and artificial intelligence (AI)

[1] An example is the restriction the European Commission imposed on Google as part of its merger with Fitbit (European Commission, 2020).

correlates positively with sales, employment, and innovation.[2] Asriyan and Kohlhas (2024) show that the increasing availability of big data to firms improves the accuracy of their expectations and significantly boosts total factor productivity. Mihet et al. (2025) study how firms' access to data and their AI adoption shape competition and give rise to market power. Adams et al. (2025) document that the stock returns of firms that adopt AI pricing are more responsive to monetary policy shocks.[3]

In contrast to all these papers, we study how the prevalence of big data shapes the propagation of cyclical fluctuations. Our theoretical model builds on the frameworks of Eeckhout and Veldkamp (2022) and Farboodi and Veldkamp (forthcoming). Eeckhout and Veldkamp (2022) analyze how data can give rise to market power and affects the measurement thereof. Farboodi and Veldkamp (forthcoming) study how data and its endogenous accumulation affect welfare, economic growth, and the measurement of GDP.[4] Data and capital are strategic complements in these models. We analyze how this complementarity affects the way in which big data shapes the transmission of cyclical fluctuations.

Second, our work is related to the macroeconomic literature on research and development (R&D) and intangible assets. Chiavari and Goraya (2022) and De Ridder (2024) show that the increasing importance of intangible inputs can account for recent trends such as the rise of market power, reduced business dynamism, and lower productivity growth. Ottonello and Winberry (2024) study how financial frictions shape the investment and innovation decisions of firms.[5] The features of data that we model are not considered in this literature on R&D and intangible assets. Finally, Aghion et al. (2023) argue that the disproportionate investment of larger firms in intangibles and information technology (IT), documented by Crouzet and Eberly (2019) and Lashkari et al. (2024), can account for secular trends in growth and rents—a pattern that chimes with our finding that data-intensive firms play an increasingly dominant role in aggregate investment.

Third, our paper is related to the literature on heterogeneity in firms' responsiveness to aggregate shocks. Previous work has focused on the relationship between firms' sensitivity to monetary policy and their size (Gertler and Gilchrist, 1994; Kroen et al., 2021), liquidity (Jeenas, 2019), default risk (Bernanke et al., 1999; Ottonello and Winberry, 2020), industry

[2]Many papers study how firms' data access correlates with firm characteristics. Examples are Begenau et al. (2018), Calderón and Rassier (2022), Corrado et al. (2022), Demirer et al. (2022), Galdon-Sanchez et al. (2022), Mukerji (2022), Quan (2022), Wu (2023), and Lashkari et al. (2024).

[3]Glocker and Piribauer (2021) present evidence that because prices are more easily adjustable in digital markets (Gorodnichenko and Talavera, 2017; Gorodnichenko et al., 2018), increases in the amount of sales that are conducted through digital retail dampen the real effects of monetary policy.

[4]Wang et al. (2022), Wu and Zhang (2022), Xie and Zhang (2022), and Ansari (2023) build on Farboodi and Veldkamp (forthcoming) and study the role of data in growth models.

[5]Acemoglu and Restrepo (2018) and Jaimovich et al. (2021) consider how automation shapes the economy.

(Durante et al., 2022), price rigidities (Meier and Reinelt, 2024), and age (Cloyne et al., 2023). Further, Crouzet and Mehrotra (2020) show that large firms are less cyclically sensitive.

Fourth, our paper relates to the research on the role of uncertainty for firm-level investment (Bloom, 2009; Bachmann et al., 2013; Kumar et al., 2022). Bloom et al. (2018) establish that firms that face higher uncertainty are less responsive to aggregate shocks. In Veldkamp (2005), Van Nieuwerburgh and Veldkamp (2006), Ordoñez (2013), and Fajgelbaum et al. (2017), there is a data feedback loop at the aggregate level, which amplifies business cycles by creating countercyclical movements in aggregate uncertainty. We build on this line of analysis by considering the possibility that firms can reduce their uncertainty endogenously.

Fifth, we contribute to the literature on incomplete information (Lucas, 1972) and rational inattention (pioneered by Sims, 2003), which establishes how agents allocate their attention and how this can account for inertia in macroeconomic outcomes. Whereas this literature focuses on the allocation of scarce attention, we study how firms' access to data and their ability to acquire data endogenously shape the propagation of cyclical fluctuations.[6] Finally, Dong et al. (2025) show that endogenous information acquisition interacts with financial constraints to amplify business cycle fluctuations. In contrast, we study how heterogeneity in firms' data access shapes their investment responsiveness to business cycle shocks and document a U-shaped relationship between data intensity and cyclicality.

2 Empirical Evidence

In this section, we construct a firm-level measure of *data intensity*, characterize how data-intensive firms differ from other firms, and establish our main empirical finding: the cyclicality of firms' investment is U-shaped in their data intensity.

2.1 Data

Measuring data intensity. To measure firms' data intensity, we use employment-biography data from Revelio Labs, encompassing the universe of public LinkedIn profiles, with nearly 500 million workers and approximately 1.8 billion employment spells. This linked employer-employee dataset includes start and end dates for each employment spell, along with a task-based classification of job roles. The job taxonomy identifies activities associated with each job title by comparing descriptions from resumés and online profiles against responsibilities listed in online job postings.[7] The classification algorithm uses workers' self-reported

[6] Our analysis in Appendix D.3 is in part related to Gondhi (2023) and Charoenwong et al. (2024), who consider models in which firms receive signals about their idiosyncratic productivity draws.

[7] For further details, see www.reveliolabs.com.

job titles, descriptions, and skills from their LinkedIn profiles to train an embeddings model that creates a mathematical representation of each position, which is then assigned to job categories using a clustering algorithm. While this approach relies on self-reported information, the public nature of LinkedIn profiles, accessible by current and former colleagues, provides a strong incentive for accurate reporting, mitigating concerns about strategic misrepresentation of job roles.

Initially, the algorithm categorizes 1,500 job roles, which are successively aggregated using a clustering algorithm. We use the taxonomy at the hierarchical level where 150 distinct roles are defined. The job titles we classify as data-related are *data analyst*, *data engineer*, *data scientist*, and *database administrator*.[8] As a robustness check, we also consider a broader measure of data-related jobs where we further include job roles classified as "business analyst" or "information specialist." Our dataset is restricted to job spells from the U.S., where we compute the firm-level count of individuals employed in data-related roles at the end of the previous year. We use sampling weights, provided by Revelio Labs, to account for the potentially non-representative nature of our sample for certain job roles and locations. Our primary measure of data intensity, $\textit{Data intensity}_{f,t-1}$, normalizes the number of workers in data-related roles by firm-level employment as recorded on LinkedIn, measured at the end of the previous year. We similarly construct a measure for the intensity of IT-related workers.[9]

Importantly, our measure captures data-related labor involved in generating, structuring, processing and analyzing data for decision-making, rather than general IT infrastructure or software maintenance. This distinction is crucial: while IT employment reflects technological capacity broadly, data-related roles are directly tied to predictive analytics and information processing, which are central to the mechanisms studied in this paper.

Other firm-level data. We merge the Revelio-based data-intensity measures with firm financials from Compustat using CIK identifiers. We exclude finance, insurance, real estate, and utilities sectors (NAICS 52, 53, and 22). Our main outcome variable is the natural logarithm of capital expenditure, $\ln(\textit{Capital expenditure})_{f,t}$.[10] As an alternative outcome variable, we use $\textit{Investment ratio}_{f,t}$, the ratio of quarterly capital expenditure to the previous quarter's capital stock (computed and winsorized as in Ottonello and Winberry, 2020). Firm-

[8] We do not use the O-NET classification as it lacks specific categories corresponding to these roles.

[9] The corresponding job roles are *application engineer*, *application support*, *devops engineer*, *information security*, *information specialist*, *infrastructure engineer*, *IT analyst*, *IT project manager*, *IT specialist*, *network specialist*, *software developer*, *software engineer*, *solutions specialist*, *systems engineer*, *technical support*, *technical support engineer*, *technology analyst*, *technology lead*, *UX designer*, and *web developer*.

[10] Compustat reports capital expenditure (*capxy*) on a year-to-date basis within the firm's fiscal year, so the raw quarterly series cumulates over the fiscal year and resets in the first fiscal quarter. We therefore de-cumulate it into a quarterly flow.

Table 1: Summary Statistics

Variable	Mean	Std. dev.	Min	Max	N
ln(Capital expenditure)	1.034	2.824	-6.908	9.900	200,113
Investment ratio	0.099	0.245	0.000	9.860	198,919
ln(Employment)	4.622	2.253	0.000	11.763	200,113
Data intensity (%)	0.743	2.064	0	100	200,113
Data intensity (alt, %)	1.456	2.984	0	100	200,113
IT employment share	0.124	0.144	0.000	1.000	200,113
Y shock	0.009	0.703	-2.611	1.736	100
I shock	0.100	0.725	-2.852	1.502	100

The table reports summary statistics for the quarterly estimation sample (firm-quarters), 1993 to 2017, excluding finance, insurance, real estate, and utilities (NAICS 52, 53, and 22). *Capital expenditure*$_{f,t}$ is the within-fiscal-year quarterly capital-expenditure flow of firm f, and *Investment ratio*$_{f,t}$ equals that flow over the previous quarter's capital stock. *Employment*$_{f,t-1}$ is the number of employees of firm f in year $t-1$. *Data intensity*$_{f,t-1}$ is the sample-weighted share of data-related employees (roles classified as "data analyst," "data engineer," "data scientist," or "database administrator") of firm f at the end of year $t-1$. *Data intensity (alt)*$_{f,t-1}$ also includes "business analyst" and "information specialist," and *IT employment share*$_{f,t-1}$ is the share of IT-related employees. *Y shock*$_t$ and *I shock*$_t$ are the quarterly real GDP and investment "main business cycle shocks" of Angeletos et al. (2020). The sample consists of firm-quarters with non-missing ln(*Capital expenditure*) and *Data intensity*. The investment ratio has fewer observations because it additionally requires the lagged capital stock (its denominator), which is missing for some firm-quarters.

level controls, such as the natural logarithm of employment, the IT employment share, leverage, the cash-to-assets ratio, the intangible-asset ratio, and firm age, are measured at annual frequency and enter with a one-year lag, so that they are predetermined relative to the contemporaneous shock.

Business cycle shocks. Our measures of aggregate fluctuations are the two "main business cycle shocks" of Angeletos et al. (2020): the shock that accounts for the bulk of business-cycle-frequency volatility in real GDP (*Y shock*$_t$) and the analogous shock for investment (*I shock*$_t$). Both are signed so that a positive realization is expansionary.

Sample and summary statistics. Because the Angeletos et al. (2020) shocks are available through 2017, our estimation sample runs from 1993 to 2017. Table 1 reports summary statistics. The distribution of data intensity is highly skewed: it has a large mass at zero—about 52% of firm-quarters have no measured data workers in our baseline estimation sample—and a thin, dispersed right tail. Our nonparametric (binned) regression approach, discussed later, is well suited to deal with such a skewed data-intensity distribution.

Table 2: Summary Statistics by Data Intensity—Annual Frequency

Panel A: High data intensity	Mean	Std. dev.	Min	Max	*N*
ln(Capital expenditure)	3.369	2.655	-6.908	10.824	15,440
Investment ratio	0.435	0.643	0.000	9.765	14,715
ln(Employment)	5.462	1.790	0.000	11.763	29,360
IT employment share	0.174	0.144	0.000	0.857	29,360
Firm age	15.656	15.667	0.000	67.000	16,090
Sales-growth volatility	0.245	0.331	0.001	4.294	14,171
Cost of debt	0.084	0.077	0.000	1.000	12,245
Panel B: Low data intensity	Mean	Std. dev.	Min	Max	*N*
ln(Capital expenditure)	1.918	2.896	-6.908	10.808	44,658
Investment ratio	0.427	0.761	0.000	10.000	41,843
ln(Employment)	3.032	2.203	0.000	10.888	117,502
IT employment share	0.104	0.173	0.000	1.000	117,502
Firm age	12.602	13.292	0.000	67.000	48,304
Sales-growth volatility	0.328	0.478	0.000	6.266	38,625
Cost of debt	0.097	0.100	0.000	1.000	34,717

Summary statistics at the firm-year level for observations with *Data intensity*$_{f,t}$ in the top quintile (Panel A) and the bottom four quintiles (Panel B) of the within-year distribution, 1993–2017, excluding finance, insurance, real estate, and utilities (NAICS 52, 53, and 22). The table is at the firm-year level because sales-growth volatility and the cost of debt are annual firm characteristics. *Firm age*$_{f,t}$ is the number of years since the firm first appears in the data; *Sales-growth volatility*$_{f,t}$ is the standard deviation of log sales growth over a trailing five-year window; *Cost of debt*$_{f,t}$ is interest expense over the two-year average of total debt.

How data-intensive firms differ. Table 2 splits firms into the top quintile of the within-year data-intensity distribution (Panel A) and the remaining four quintiles (Panel B). High-data-intensity firms invest more and are larger, leaving their investment ratios similar, and have more IT-related employees than those with lower data intensity. High-data-intensity firms are furthermore older (15.7 versus 12.6 years), exhibit less volatile sales growth (a standard deviation of 0.25 versus 0.33), and face a lower cost of debt (a ratio of interest expenses over the two-year average of total debt of 0.08 versus 0.10). These patterns are consistent with the model we set up later, in which superior access to data reduces the variance of firms' productivity and lowers the cost of investing.

2.2 Empirical Strategy

To flexibly estimate how the cyclicality of investment varies with (lagged) data intensity, we estimate this relationship nonparametrically by partitioning firms into bins and interacting each bin with the business cycle shocks. We first define these bins by using fixed percentage-

point bins of data intensity in $t-1$: $\{0\}, (0, 0.5], (0.5, 1], (1, 2], (2, 4], (4, 8], (8, \infty)$. This definition can directly inform how business cycle shocks affect investment as data intensity increases overall by moving to a higher bin. The drawback of this approach, however, is that trends in data intensity might be correlated with other trends. Therefore, we later turn to a definition of the bins that is based on the within-year distribution of data intensity: in each year, firms with no data workers form one bin (DI=0), and the firms with positive data intensity are sorted into within-year quintiles, $Q1$ (least data-intensive) through $Q5$ (most data-intensive). In both cases, our nonparametric approach is essential to uncover potential nonlinearities in the relationship between investment cyclicality and data intensity.

We estimate two complementary specifications. The first one, which we refer to as the "absolute specification," interacts the shock with every bin indicator:

$$y_{f,t} = \sum_b \beta_b \, \mathbb{1}\{f \in b\}_{t-1} \, s_t + \sum_b \gamma_b \, \mathbb{1}\{f \in b\}_{t-1} + X'_{f,t-1}\theta + \alpha_f + \delta_{j(f),\tau(t)} + \varepsilon_{f,t}, \qquad (1)$$

where $y_{f,t} = \ln(\textit{Capital expenditure}^q_{f,t})$, $\mathbb{1}\{f \in b\}_{t-1}$ indicates that firm f's lagged data-intensity bin is b, s_t is one of the Angeletos et al. (2020) shocks, $X_{f,t-1}$ collects the lagged firm controls (lagged ln(employment) and the lagged IT employment share in our baseline), α_f are firm fixed effects, and $\delta_{j(f),\tau(t)}$ are industry-by-year fixed effects (four-digit NAICS $\times$ year). Because each firm-quarter belongs to exactly one bin, $\beta_b = \partial y_{f,t}/\partial s_t$ is the *level* of investment cyclicality of bin b. Identification of these levels requires that the shock is not absorbed by the time fixed effects: at quarterly frequency the shock varies *within* the calendar year, so industry-by-year fixed effects absorb time-varying unobserved heterogeneity at the industry level and seasonal composition. The four within-year quarterly shock realizations then identify the coefficients $\{\beta_b\}$.

The second specification, which we refer to as the "relative specification," drops the DI=0 interaction and saturates the time dimension:

$$y_{f,t} = \sum_{b \neq 0} \beta_b \, \mathbb{1}\{f \in b\}_{t-1} \, s_t + \sum_b \gamma_b \, \mathbb{1}\{f \in b\}_{t-1} + X'_{f,t-1}\theta + \alpha_f + \delta_{j(f),t} + \varepsilon_{f,t}, \qquad (2)$$

where $\delta_{j(f),t}$ are industry-by-year-quarter fixed effects, which absorb the shock entirely along with any other aggregate or industry-time confounder. Here β_b measures the *differential* cyclicality of bin b relative to the omitted DI=0 bin. The two specifications are complementary: (1) delivers the *levels* of cyclicality and hence the overall shape, which is why we use it for our headline figure, while (2) is the more demanding design that identifies the cross-sectional shape from purely within-industry-quarter variation. Throughout, we report standard errors two-way clustered by firm and year-quarter.

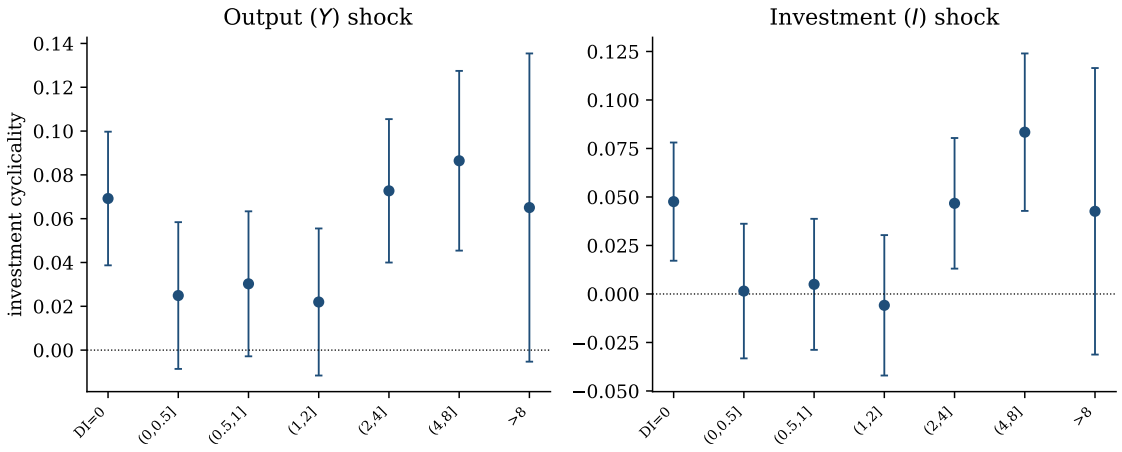
Table 3: Investment Cyclicality with Fixed Percentage-Point Bins

	Absolute		Relative	
	Y shock	I shock	Y shock	I shock
DI=0	0.069***	0.048**		
	(0.019)	(0.019)		
(0,0.5]	0.025	0.001	-0.045***	-0.048***
	(0.020)	(0.021)	(0.017)	(0.018)
(0.5,1]	0.030	0.005	-0.039**	-0.045***
	(0.020)	(0.021)	(0.016)	(0.017)
(1,2]	0.022	-0.006	-0.052***	-0.060***
	(0.020)	(0.022)	(0.019)	(0.020)
(2,4]	0.073***	0.047**	0.003	-0.005
	(0.020)	(0.020)	(0.022)	(0.022)
(4,8]	0.086***	0.083***	0.020	0.037
	(0.025)	(0.025)	(0.030)	(0.029)
>8	0.065	0.043	0.004	-0.005
	(0.043)	(0.045)	(0.049)	(0.050)
Dip: $(1,2]-$(DI=0)	-0.047***	-0.053***	-0.052***	-0.060***
	(0.017)	(0.019)	(0.019)	(0.020)
Rebound: $(4,8]-(1,2]$	0.064**	0.089***	0.072**	0.097***
	(0.027)	(0.027)	(0.030)	(0.030)
$(4,8]-$(DI=0)	0.017	0.036	0.020	0.037
	(0.025)	(0.025)	(0.030)	(0.029)
Firm FE	Y	Y	Y	Y
Industry×year FE	Y	Y	N	N
Industry×year-quarter FE	N	N	Y	Y
Observations	199,901	199,901	193,809	193,809

The sample is 1993–2017 (firm-quarters). The dependent variable is $\ln(\textit{Capital expenditure}^{q}_{f,t})$, the quarterly capital-expenditure flow. Firms with no data workers form the DI=0 bin; the remaining firms are sorted into fixed percentage-point bins of $\textit{Data intensity}_{f,t-1}$—$(0, 0.5], (0.5, 1], (1, 2], (2, 4], (4, 8], (8, \infty)$—and each bin indicator is interacted with the shock. The shocks are the real GDP (Y) and investment (I) "main business cycle shocks" of Angeletos et al. (2020). "Absolute" columns report the bin-specific cyclicalities β_b from (1) (each is a bin's response of log capital expenditure to a one-unit positive shock); "Relative" columns report differential cyclicalities from (2) relative to the omitted DI=0 bin. *Dip* is $(1, 2]-$(DI=0); *Rebound* is $(4, 8]-(1, 2]$, the analogues of the quintile contrasts in Table 4. All columns include lagged *ln(Employment)* and the lagged *IT employment share* and firm fixed effects; "Absolute" columns add industry-by-year fixed effects (leaving the shock in, so coefficients are cyclicality levels), and "Relative" columns add industry-by-year-quarter fixed effects (which absorb the shock, so coefficients are differences relative to DI=0). Standard errors, two-way clustered by firm and year-quarter, are in parentheses.

Local projections. To capture the dynamic effects of business cycle shocks, we estimate the same specifications at horizons $h = 0, 1, \ldots, 12$ quarters by replacing the dependent variable with its h-quarter lead $y_{f,t+h}$ (Jordà, 2005). By construction, the $h = 0$ projection reproduces the impact regression. We focus here on the within-year bin definition.

Figure 2: Investment Cyclicality with Fixed Percentage-Point Bins



Bin-specific investment cyclicalities at impact, β_b, from the absolute specification with fixed percentage-point bins of *Data intensity*$_{f,t-1}$, for the *Y shock* (left) and *I shock* (right). Each point is a bin's response of log capital expenditure to a one-unit positive shock; whiskers are 90% confidence intervals from standard errors two-way clustered by firm and year-quarter. Sample: 1993–2017.

2.3 Results

Table 3 and Figure 2 show the results for our fixed-bin specification. Investment is highly non-monotonic in firms' data intensity. Investment cyclicality dips for firms in the low and intermediate fixed percentage-point bins, and it rises again for the most data-intensive firms, producing a U-shape, as Figure 2 shows graphically. Table 3 shows that the dip at the low bins and the rebound at high data intensities are both statistically significant across shocks and specifications.

As mentioned above, this fixed-bin ranking has the drawback that data intensity has risen over our sample period. We therefore now focus on a within-year ranking of data intensity, which is robust to this trend.

Table 4 reports the impact estimates based on both (1) and (2) for the within-year definition of data-intensity bins. Again, three patterns stand out and recur across both shocks. First, investment is procyclical throughout the distribution: no bin displays negative cyclicality, and the lowest bin is significantly procyclical (a one-unit *Y shock* raises capital expenditure by 6.9% on impact). Second, cyclicality falls across the lower positive quintiles: in the relative specification (columns 3 and 4), firms in $Q1$ are significantly less cyclical than zero-intensity firms ($Q1 - \text{DI=0} = -0.041$, s.e. 0.018 for the *Y shock*), and $Q1$ through $Q4$ form a flat, depressed plateau. Third, cyclicality rises sharply at the top: the most data-intensive quintile is significantly more cyclical than the trough ($Q5 - Q1 = 0.044$, s.e. 0.017) and recovers to approximately the cyclicality of the zero-intensity firms. A joint test

Table 4: Investment Cyclicality and Data Intensity

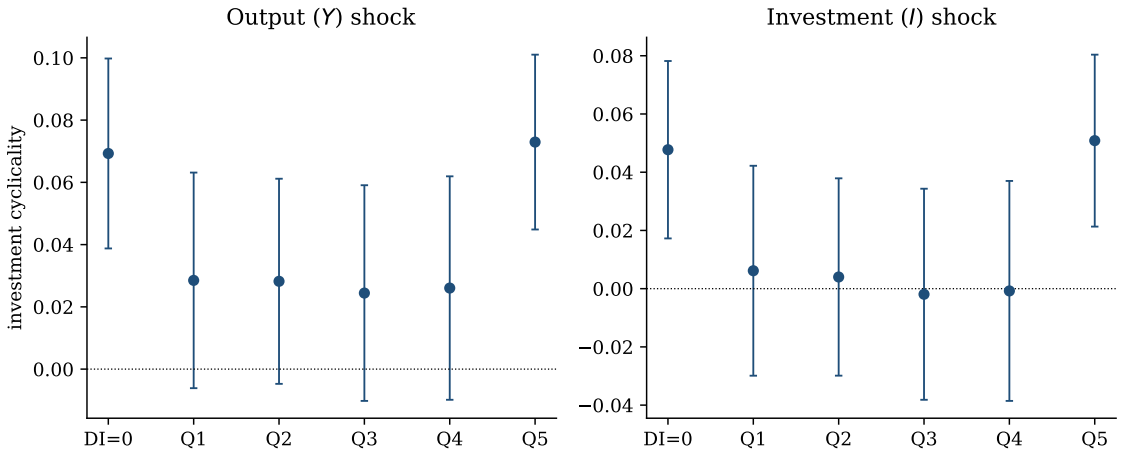
	Absolute		Relative	
	Y shock	I shock	Y shock	I shock
DI=0	0.069***	0.048***		
	(0.019)	(0.019)		
Q1	0.029	0.006	-0.041**	-0.044**
	(0.021)	(0.022)	(0.018)	(0.018)
Q2	0.028	0.004	-0.039**	-0.041**
	(0.020)	(0.021)	(0.016)	(0.017)
Q3	0.024	-0.002	-0.046**	-0.054***
	(0.021)	(0.022)	(0.019)	(0.019)
Q4	0.026	-0.001	-0.049**	-0.056***
	(0.022)	(0.023)	(0.020)	(0.021)
Q5	0.073***	0.051***	0.002	-0.001
	(0.017)	(0.018)	(0.020)	(0.021)
Dip (single): Q1 − DI=0	-0.041**	-0.042**	-0.041**	-0.044**
	(0.016)	(0.018)	(0.018)	(0.018)
Dip (pooled): $\overline{Q1–Q4}$−DI=0	-0.042***	-0.046***	-0.044***	-0.049***
	(0.015)	(0.016)	(0.016)	(0.016)
Rebound: Q5 − Q1	0.044***	0.045***	0.044**	0.043**
	(0.017)	(0.017)	(0.020)	(0.020)
Q5 − DI=0	0.004	0.003	0.002	-0.001
	(0.017)	(0.018)	(0.020)	(0.021)
Joint test (equal cyclicality), p	0.007	0.006	–	–
Firm FE	Y	Y	Y	Y
Industry×year FE	Y	Y	N	N
Industry×year-quarter FE	N	N	Y	Y
Observations	199,901	199,901	193,809	193,809

The sample is 1993–2017 (firm-quarters). The dependent variable is $\ln(\textit{Capital expenditure}^{q}_{f,t})$, the quarterly capital-expenditure flow. Firms with no data workers form the DI=0 bin; the remaining firms are sorted into within-year quintiles $Q1, \ldots, Q5$ of (predetermined) data intensity, and each bin indicator is interacted with the shock. The shocks are the real GDP (Y) and investment (I) "main business cycle shocks" of Angeletos et al. (2020). "Absolute" columns report the bin-specific cyclicalities β_b from (1) (each is a bin's response of log capital expenditure to a one-unit positive shock); "Relative" columns report differential cyclicalities from (2) relative to the omitted DI=0 bin. *Dip (single)* is $Q1-$(DI=0); *Dip (pooled)* is the average of $Q1$–$Q4$ minus DI=0; *Rebound* is $Q5-Q1$. The *Joint test* reports the p-value of the hypothesis that all bins share a common cyclicality (reported only for the absolute specification, where the levels are identified). All columns include lagged *ln(Employment)* and the lagged *IT employment share* and firm fixed effects; "Absolute" columns add industry-by-year fixed effects (leaving the shock in, so coefficients are cyclicality levels), and "Relative" columns add industry-by-year-quarter fixed effects (which absorb the shock, so coefficients are differences relative to DI=0). Standard errors, two-way clustered by firm and year-quarter, are in parentheses.

that cyclicality is equal across all bins is rejected. Figure 3 illustrates this U-shaped pattern graphically, plotting the cyclicality levels from the absolute specification (1).

Although these bins are defined by firms' rank within each year $t-1$, they remain

Figure 3: Investment Cyclicality by Data-Intensity Bin

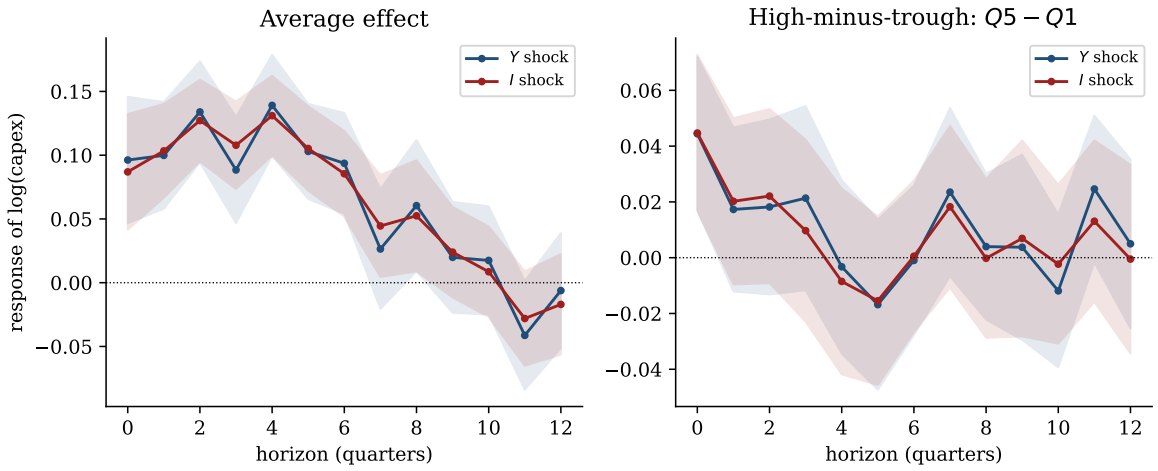


Bin-specific investment cyclicalities $\hat{\beta}_b$ from the absolute specification (1) at impact, for the *Y shock* (left) and the *I shock* (right). Each point is a bin's response of log capital expenditure to a one-unit positive shock. Bins are the zero-intensity firms (DI=0, the leftmost point) and within-year quintiles $Q1, \ldots, Q5$ of the positive-intensity distribution. Whiskers are 90% confidence intervals from standard errors two-way clustered by firm and year-quarter. The corresponding estimates are reported in Table 4. Sample: 1993–2017.

informative about the relationship between absolute data access and investment cyclicality. Firms with very low data intensity are present in every year and, thus, provide a common absolute benchmark. Moreover, as overall data intensity rises over the sample period, a given positive-intensity quintile corresponds to progressively higher absolute levels of data access. Early cross-sections may therefore primarily identify the downward-sloping part of the relationship, whereas later cross-sections increasingly span the upward-sloping part as well. The fixed percentage-point bins of Table 3, which hold absolute thresholds constant across years, deliver the same pattern, confirming that the within-year results do not merely reflect firms' relative standing.

The average effect of both shocks itself is positive in the first two years (Figure 4, left panel). Estimating the bin specification at horizons $h = 0, \ldots, 12$ shows that the U-shape is concentrated at short horizons: the high-minus-trough differential $\hat{\beta}^h_{Q5} - \hat{\beta}^h_{Q1}$ is positive and significant on impact, remains positive for up to the fourth quarter, and then dissipates as the average response mean-reverts (Figure 4, right panel). The interaction with high data intensity is economically substantial as the $\hat{\beta}^h_{Q5} - \hat{\beta}^h_{Q1}$ differential is about half as large as the average impact effect.

Figure 4: Dynamics: Average Effect and the High-Minus-Trough Differential



Local projections at horizons $h = 0, \ldots, 12$ quarters. Left: the average response of log capital expenditure to a one-unit positive business cycle shock. Right: the high-minus-trough differential $\hat{\beta}^h_{Q5} - \hat{\beta}^h_{Q1}$ from the absolute specification. Shaded bands are 90% confidence intervals from standard errors two-way clustered by firm and year-quarter.

2.4 Robustness

Aggregate shocks may transmit to firm-level investment through several channels: for example, financial frictions that depend on leverage and liquidity, the size and age dependence of financial constraints, and the role of asset intangibility. As Table 2 shows, data intensity is correlated with these characteristics: data-intensive firms are larger, older, and face lower cost of debt. The U-shape could therefore in principle reflect the differential cyclicality that these other channels induce, rather than a role for data *per se* if data intensity is acting as a proxy for them.

We address this directly by re-estimating regression specification (1) and additionally interacting the shock with lagged firm size (logged employment), leverage, the cash-to-assets ratio, firm age, and the intangible-asset ratio (as data can be viewed as a subcategory of intangible assets). Each of these interactions allows the corresponding channel to carry its own shock-cyclicality, so that the data-intensity bins capture only the component of investment cyclicality that is orthogonal to these other channels.

Panel A of Table 5 shows the results. The entire cyclicality profile is essentially unchanged. In particular, the high-data-intensity rebound retains its magnitude and significance (the rebound $Q5 - Q1$ moves from 0.044 to 0.036 for the Y *shock* and from 0.045 to 0.036 for the I *shock*, both statistically significant). The point estimates for the dip from firms with $DI = 0$ to firms in Q1 become slightly weaker, but remain negative and statis-

Table 5: Investment Cyclicality and Data Intensity—Robustness

	Absolute		Relative	
	Y shock	I shock	Y shock	I shock
Panel A: baseline, controlling for transmission channels				
Dip (pooled): lower bins − DI=0	-0.023*	-0.028*	-0.024*	-0.029*
	(0.013)	(0.014)	(0.014)	(0.015)
Rebound: top − lowest positive	0.036**	0.036**	0.039**	0.041**
	(0.016)	(0.017)	(0.020)	(0.020)
Observations	174,526	174,526	168,417	168,417
Panel B: broader data-intensity measure				
Dip (pooled): lower bins − DI=0	-0.044***	-0.047***	-0.045***	-0.049***
	(0.014)	(0.016)	(0.015)	(0.016)
Rebound: top − lowest positive	0.039**	0.039**	0.033*	0.032*
	(0.016)	(0.016)	(0.019)	(0.019)
Observations	199,901	199,901	193,809	193,809
Panel C: investment ratio				
Dip (pooled): lower bins − DI=0	-0.003*	-0.003**	-0.004**	-0.004**
	(0.002)	(0.002)	(0.002)	(0.002)
Rebound: top − lowest positive	0.007**	0.008***	0.006*	0.007**
	(0.003)	(0.002)	(0.003)	(0.003)
Observations	205,747	205,747	199,646	199,646
Panel D: annual frequency				
Dip (pooled): lower bins − DI=0	-0.133**	-0.120**	-0.120**	-0.116**
	(0.056)	(0.054)	(0.059)	(0.051)
Rebound: top − lowest positive	0.117**	0.100*	0.140***	0.116**
	(0.057)	(0.060)	(0.052)	(0.052)
Observations	58,091	58,091	56,809	56,809

Notes. The sample is 1993–2017. "Dip (pooled)" is the average of the lower bins minus the zero-data bin ($\overline{Q1–Q4}$ − (DI=0)), and "Rebound" is the top bin minus the lowest positive bin ($Q5-Q1$). All specifications include firm fixed effects and lagged ln(employment) and the lagged IT employment share. *Absolute* columns add industry-by-year fixed effects, leave the shock in the regression, and report the cyclicality *level* of each bin; *Relative* columns add industry-by-year-quarter fixed effects (industry-by-year fixed effects in the annual panel), so the coefficients are *differences* in cyclicality relative to the omitted DI=0 bin. *Panel A* adds to the baseline specification interactions of the shock with five lagged firm characteristics: ln(employment) (firm size), leverage, the cash-to-assets ratio, firm age, and the intangible-asset ratio. *Panel B* broadens the data-intensity measure to also count business analysts and information specialists. *Panel C* uses the investment ratio as the dependent variable. *Panel D* estimates the relation at annual frequency; there the absolute specification controls for lagged CPI inflation, the unemployment rate, and real GDP growth (as well as year instead of industry-by-year fixed effects), and the relative specification uses industry-by-year fixed effects. Standard errors, two-way clustered by firm and year-quarter (by firm and year in the annual panel), are in parentheses.

tically significant. These results suggest that the cyclicality of highly data-intensive firms' investment is not an artifact of those firms' size, age, leverage, liquidity, or asset intangibility.

Data intensity operates over and above the channels through which aggregate fluctuations conventionally reach investment.

Panel B of Table 5 shows that our results are robust when using a broader measure of data intensity (including business analysts and information specialists). Panel C shows the same pattern when using the investment ratio instead of capital expenditures as the dependent variable, and Panel D shows the results when focusing on annual frequency.

Appendix A further shows that our results are robust to different control specifications. Further, Appendix Table A.2 reports additional robustness checks. First, the U-shape is specific to investment. When we use ln(sales) as the dependent variable, sales respond positively to business cycle shocks in all bins, but there is neither a dip nor a rebound (Panel A). Hence, our results do not merely reflect a differential exposure of data-intensive firms to demand. Second, the U-shape is not driven by firms with very few LinkedIn profiles or by poor LinkedIn coverage. The results are similar when we drop firms with fewer than 20 LinkedIn profiles or fewer than 20 employees in Compustat (Panels B and C), when we measure data intensity without sampling weights (Panel D), and when we split the DI=0 bin by LinkedIn coverage (Panel E).

Next, we want to understand the underlying mechanisms of this U-shaped pattern. We therefore now turn to our theoretical framework.

3 Model

Production & investment. There is a unit mass of infinitely-lived firms, indexed by $i \in [0, 1]$. Time is discrete and denoted by $t = 0, 1, 2, ...$ The production function is

$$Y_{i,t} = A_{i,t} K_{i,t}^{\alpha}, \tag{3}$$

where $\alpha \in (0, 1)$, $Y_{i,t}$ denotes the output produced by firm i in period t, $A_{i,t}$ the firm's productivity, and $K_{i,t}$ its capital stock. Every firm i chooses the capital stock $K_{i,t+1}$ in period t before observing $A_{i,t+1}$.[11] A firm's investment in a given period t is denoted by $I_{i,t} = K_{i,t+1} - (1 - \delta)K_{i,t}$, where $\delta \in [0, 1]$ is the depreciation rate.

Data & productivity. Superior access to data favorably affects a firm's productivity distribution in two possible ways, namely by (i) increasing the firm's expected productivity and (ii) by reducing the variance of its productivity. The quality of the data a firm utilizes in period t is represented by the object $\sigma_{i,t}$, where a smaller $\sigma_{i,t}$ represents better access to

[11] Including labor in the production function does not alter our analysis if it is hired on the spot market after productivity is revealed.

data. One can view $\sigma_{i,t}$ as the noise of a signal the firm receives about a productivity-relevant state, where a firm with superior access to data receives a more precise (less noisy) signal.

Conditional on the information set in period t, a firm's expected productivity in period $t+j$, with $j \geq 1$, is given by:

$$\mathbb{E}_t[A_{i,t+j}] = \bar{A}_{t+j} - \kappa_e \sigma_{i,t+j}, \tag{4}$$

where $\bar{A}_{t+j}$ is the aggregate productivity component that affects all firms equally. The variance of a firm's productivity (conditional on all information available in period t) is

$$VAR_t[A_{i,t+j}] = \bar{V}_{t+j} + \kappa_v \sigma_{i,t+j}. \tag{5}$$

The parameters $\kappa_e \geq 0$ and $\kappa_v \geq 0$ capture the effects of data on a firm's expected productivity and its variance. These relationships arise endogenously if the firm matches an unknown payoff-relevant state such as an optimal marketing approach using data (Farboodi and Veldkamp, forthcoming). We assume that these relationships are linear for analytical tractability. Later, we also discuss decreasing and increasing returns to scale in data.

Investment cost. Per unit of investment $I_{i,t}$, a firm pays the cost $C(r_t, \rho_i, \sigma_{i,t+1})$, where

$$C(r_t, \rho_i, \sigma_{i,t+1}) = r_t + \rho_i VAR_t[A_{i,t+1}], \tag{6}$$

and r_t is an aggregate interest rate that is the same for every firm. We assume that the monetary authority keeps this real rate constant, $r_t = r$ at all $t \geq 0$. We assume that $\rho_i \geq 0$, i.e., that firms which face lower uncertainty have lower investment costs. This reflects the risk-return relationship (an asset's risk is positively correlated with its expected return). Firms with superior access to data thus face lower investment costs because the variance of their productivity is smaller. Later, we discuss why this specification can be viewed as a reduced-form representation of managers' risk aversion.

Data sources. The quality of the data a firm has access to in a given period is based on three components, namely (i) how much data the firm exogenously has access to, (ii) how much data the firm chooses to directly acquire (at a cost), and (iii) how much data the firm acquires as a by-product of production through a data feedback loop. Formally, a firm's access to data is represented by the following functional form:

$$\sigma_{i,t+1} = \bar{\sigma}_i - \eta_i d_{i,t+1} - \zeta_i K_{i,t+1}. \tag{7}$$

For every firm i, the value of $\bar{\sigma}_i$ is exogenously given. The amount of data a firm chooses to acquire for utilization in period $t+1$ is denoted by $d_{i,t+1}$. The parameter $\eta_i \geq 0$ governs the extent to which a firm's access to data can be improved through direct acquisition of data. The costs of data acquisition are given by $\mathcal{C}(d_{i,t+1}) = (d_{i,t+1})^2$. If $\zeta_i > 0$, there is an active data feedback loop (as in Farboodi and Veldkamp, forthcoming). Then, data accumulates as a by-product of production, e.g., because a firm which produces more learns more about productivity-relevant states such as its customers' preferences. For tractability, we formulate the data feedback loop directly in terms of capital. The core economic insights extend if we specify that data is generated through production rather than capital directly. The latter creates a simultaneity issue with respect to productivity, which we circumvent with our specification.

Profits. In every period t, a firm's optimization problem is given by:

$$\max_{\{K_{i,j},d_{i,j}\}_{j=t+1}^{\infty}} \mathbb{E}_t \sum_{j=t+1}^{\infty} \beta^{j-t} \Pi_{i,j}, \tag{8}$$

where $\beta \in (0,1]$ is the discount factor and the flow profits $\Pi_{i,t}$ are, for any t, given by:

$$\Pi_{i,t} = A_{i,t}(K_{i,t})^{\alpha} - C(r_{t-1}, \rho_i, \sigma_{i,t}) I_{i,t-1} - (d_{i,t})^2. \tag{9}$$

We impose the following two assumptions which make the firm's problem well-behaved:

Assumption 1. *We assume that:*

- *The expected flow profits are strictly concave in capital, i.e.,* $\frac{\partial^2 \mathbb{E}_t[\Pi_{i,t+1}]}{\partial K_{i,t+1}^2} < 0$ *holds for every* t *and* i.
- *For any possibly optimal* $K_{i,t+1}$ *and* $d_{i,t+1}$, $\mathbb{E}_t[A_{i,t+1}]$ *and* $\sigma_{i,t+1}$ *remain strictly positive.*

Timing, equilibrium, and aggregate shocks. In every period t, any firm i chooses $K_{i,t+1}$ and $d_{i,t+1}$. When doing so, the firm knows the values of κ_e, κ_v, ρ_i, $\bar{\sigma}_i$, η_i, and ζ_i, and the future paths $\{\bar{A}_j\}_{j \geq t+1}$ and $\{\bar{V}_j\}_{j \geq t+1}$. We focus for now on a partial equilibrium setting: Every firm maximizes its profits, conditional on the idiosyncratic and aggregate states. Later, we extend the analysis to general equilibrium.

To study the cyclicality of data-intensive firms, we focus on the effects of aggregate productivity shocks on firm investment. In a given period t, this relationship is characterized by the marginal effect of a change in $\bar{A}_{t+1}$ on the firm's optimal capital choice, holding the paths $\{\bar{A}_j\}_{j \geq t+2}$, and $\{\bar{V}_j\}_{j \geq t+1}$ fixed.

Connection to the empirical analysis. In the empirical analysis, we focus on lagged data intensity, i.e., based on the share of data-related employees at the end of the previous year. This variable serves as a proxy for the predetermined access to data a firm has when the shock hits. In our baseline model, predetermined heterogeneity in firms' access to data is captured by differences in $\bar{\sigma}_i$. We therefore interpret differences in lagged data intensity as empirical counterparts to heterogeneity in $\bar{\sigma}_i$. The theoretical counterparts of the business cycle shocks we consider in the empirical analysis are the transitory shocks to $\bar{A}_{t+1}$.

Within our main theoretical analysis, $\bar{\sigma}_i$ is exogenously given for any firm and time invariant. In practice, heterogeneity in lagged data intensity may also reflect endogenous and time-varying factors, e.g., how many data scientists a firm hired in the past. The key relationship we study, namely how heterogeneity in firms' predetermined access to data affects their responsiveness to aggregate shocks, does not depend qualitatively on whether this heterogeneity arises from exogenous factors or from past endogenous choices.

We demonstrate this in Section D.2 of the Online Appendix, which presents an extension in which data accumulates over time and a firm's predetermined access to data is thus an endogenous state variable. Hence, modeling heterogeneity in predetermined data access through differences in $\bar{\sigma}_i$ provides a tractable representation of this heterogeneity without imposing relevant implicit restrictions on its underlying sources. By these arguments, the fact that our empirically estimated effects are identified off time-varying heterogeneity in lagged data intensity (in the presence of firm fixed effects) is also consistent with our theoretical approach.

Discussion of modeling assumptions and extensions. The fact that firms with better access to data have a higher expected productivity and a lower variance of productivity is empirically well-documented (Bajari et al., 2019; Corrado et al., 2022; Wu, 2023). Farboodi and Veldkamp (forthcoming) offer a microfoundation of these relationships.

Suppose that a firm's productivity $A_{i,t}$ satisfies $A_{i,t} = \bar{A}_t - d(a_{i,t}, \theta_{i,t}) + \epsilon_{i,t}$ and depends on (i) the aggregate productivity level $\bar{A}_t$, (ii) an idiosyncratic shock $\epsilon_{i,t}$, and (iii) how well the firm matches an unknown payoff-relevant state $\theta_{i,t}$ through choice of $a_{i,t}$. The function $d(\cdot)$ is some distance metric. The state $\theta_{i,t}$ can be understood as the optimal product variety or marketing approach, which the firm wishes to mirror. Every firm i observes an unbiased signal with variance $\sigma_{i,t}$ about $\theta_{i,t}$. A firm with access to superior data (i.e., a firm which observes a signal with a lower $\sigma_{i,t}$) can match $\theta_{i,t}$ more precisely. This reduces the expected value of $d(a_{i,t}, \theta_{i,t})$ and makes high values thereof less likely. Thus, reductions of $\sigma_{i,t}$ raise the expected productivity and reduce its variance.

We discuss a number of other microfoundations as well as extensions in Section 4.5. In

particular, we show that our insights extend (i) if firms face identical investment costs, but firm managers are risk averse as in Eeckhout and Veldkamp (2022), (ii) if data accumulates endogenously over time, (iii) if data does not favorably affect firms' productivity distribution but only enables firms to predict their future idiosyncratic productivities, and (iv) if data has increasing or decreasing returns to scale. In Section 5, we extend our analysis to a general equilibrium setting.

4 Analytical Results

4.1 Preliminaries

In the following, we study the impact of aggregate productivity shocks on the optimal capital stock of firms, relegating all proofs to Section B in the Online Appendix. We define the elasticity of a firm's optimal capital choice $K^*_{t+1}(\bar{\sigma}_i, \eta_i, \zeta_i)$ with respect to a change in aggregate productivity $\bar{A}_{t+1}$ as $\phi(\bar{\sigma}_i, \eta_i, \zeta_i)$. We express the optimal capital choice and the elasticity as functions of $\bar{\sigma}_i, \eta_i$, and ζ_i explicitly because we focus on the role of these data-related parameters. Note that:

$$\phi(\cdot) := \frac{\partial K^*_{t+1}(\cdot)}{\partial \bar{A}_{t+1}} \frac{\bar{A}_{t+1}}{K^*_{t+1}(\cdot)} > 0. \tag{10}$$

An increase in aggregate productivity increases the optimal capital choice of any firm. Thus, a firm responds more strongly to an aggregate productivity shock if $\phi(\cdot)$ is *larger*. We start by examining how differences in firms' data access affect their responses to aggregate productivity shocks when any firm's data access is exogenously given. Thereafter, we show how endogenous data acquisition and the data feedback loop affect these results.

4.2 Exogenous Access to Data

In this section, we consider a data economy in which there is no data feedback loop and in which firms cannot acquire data. Formally, we set $\eta_i = 0$ and $\zeta_i = 0$ for all firms i. The following proposition characterizes how data access shapes firms' responses to aggregate shocks in this benchmark—to understand these results, recall that firms with a higher $\bar{\sigma}_i$ have *worse* predetermined data access.

Proposition 1. *Suppose that $\eta_i = 0$ and $\zeta_i = 0$ for all i. Then $\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i} \geq 0$, with strict inequality if $\kappa_e > 0$, i.e., firms with better data respond less strongly to aggregate productivity shocks.*

The intuition which underlies this result is as follows: If firms with better access to data have a larger expected productivity (i.e., if $\kappa_e > 0$), these firms respond *less strongly* to aggregate productivity shocks. This is because any change of $\bar{A}_{t+1}$ will induce a relatively small percentage change in the expected productivity of firms with access to better data, so these firms respond less strongly to changes in $\bar{A}_{t+1}$. Of course, this comparative static follows directly from the positive impact of data on expected productivity. Importantly, however, we show that the negative relationship between data access and firms' responsiveness to aggregate productivity shocks also emerges if better data does not alter a firm's productivity distribution, but only enables firms to forecast the future realizations of their productivity—we formalize this in Section D.3 of the Online Appendix.

The predictions we obtain in this benchmark can rationalize the negative relationship between lagged data intensity and investment cyclicality among firms with relatively low data intensity. However, it cannot explain why the relationship between lagged data intensity and investment cyclicality is positive among firms with a relatively high data intensity.

4.3 Direct Acquisition of Data

We now study how the presence of data markets affects the propagation of business cycle shocks. Formally, we assume that $\eta_i > 0$, which means that firms can directly acquire data. To facilitate the derivation of analytical results, we focus on a benchmark in which capital depreciates fully between periods (i.e., $\delta = 1$) and in which the data feedback loop is inactive (i.e., $\zeta_i = 0$ for all firms). We refer to this benchmark as the *data markets benchmark*. We also present numerical analysis which demonstrates that our insights extend if we drop these simplifying assumptions.

We refer to the optimal amount of data a firm chooses to acquire for utilization in period $t+1$ as $d^*_{t+1}(\cdot)$. We begin by characterizing the optimal data and capital choices of firms.

Lemma 1. *Consider the data markets benchmark. A firm's optimal data acquisition choice is given by*

$$d^*_{t+1}(\cdot) = 0.5\eta_i\big[\kappa_e(K_{i,t+1})^{\alpha} + \rho_i\kappa_v K_{i,t+1}\big]. \tag{11}$$

A firm's optimal capital choice must satisfy the optimality condition

$$\alpha\big(\bar{A}_{t+1} - \kappa_e(\bar{\sigma}_i - \eta_i d^*_{t+1})\big)\,(K^*_{t+1})^{\alpha-1} = \big(r_t + \rho_i\bar{V}_{t+1} + \rho_i\kappa_v(\bar{\sigma}_i - \eta_i d^*_{t+1})\big). \tag{12}$$

Lemma 1 establishes that data and capital are strategic complements: The data optimality condition given in equation (11) formalizes that, when a firm uses more capital in a

given period, it also optimally acquires more data for utilization in that period. Intuitively, the value of acquiring better data increases with the amount of capital employed. This is because improvements in expected productivity and reductions in investment costs (attained through superior data) apply to each unit of capital, which makes data acquisition more attractive for larger firms. In addition, the capital optimality condition given in equation (12) demonstrates that access to better data strengthens a firm's investment incentives by the converse logic. The results of Lemma 1 also imply that firms with superior predetermined data access acquire more data on data markets (because these firms hold more capital). Thus, the presence of data markets further amplifies the firms' heterogeneity in data access.

The complementarity between data and capital generates endogenous amplification of aggregate productivity shocks—firms respond more strongly to these shocks when they can acquire data than when they cannot. In particular, an aggregate productivity shock has an additional second-round effect on any firm's optimal capital choice when firms can directly acquire data. To see this, consider the effects of a positive shock: As a first-round effect, this induces any firm to acquire more capital. Given that data and capital are strategic complements, the first-round effect induces firms to acquire more data and, in turn, even more capital—this is the second-round effect. Crucially, the magnitude of the second-round effect is larger for firms with superior predetermined access to data. This is because these firms are larger, so an aggregate shock induces such firms to adjust their capital stock relatively strongly, thereby leading to a relatively large change in the amount of data they acquire.

The following proposition formalizes how heterogeneity in predetermined data access shapes firms' responsiveness to these shocks when firms can acquire data directly. To facilitate the analytical derivations, we set $\alpha = 0.5$, but later show that the qualitative insights extend more generally using numerical analysis.

Proposition 2. *Consider the data markets benchmark and suppose that* $\alpha = 0.5$, $\kappa_e > 0$, *and* $\kappa_v > 0$. *Then:*

- *The relationship between a firm's predetermined data access and its responsiveness to an aggregate productivity shock is convex in data access, i.e.,* $\frac{\partial^2 \phi(\cdot)}{\partial \bar{\sigma}_i^2} > 0$.
- *Firms with better data respond less strongly to aggregate productivity shocks if and only if* $\frac{\partial \phi_d(\cdot)}{\partial \bar{\sigma}_i} > 0$, *where* $\phi_d(\cdot) := -\frac{1}{4}\big(\bar{A} - \kappa_e \bar{\sigma}_i\big) + \frac{3(\eta_i)^2 \kappa_e \kappa_v \rho_i}{8}(K^*_{t+1}) + \frac{(\eta_i \kappa_v \rho_i)^2}{2}(K^*_{t+1})^{1.5}$. *Moreover,* $\frac{\partial^2 \phi_d(\cdot)}{\partial \bar{\sigma}_i^2} > 0$.

The two results of Proposition 2 imply that the relationship between firms' predetermined data access and their responsiveness to aggregate productivity shocks is negative among firms

with poor access to data, and becomes less negative as data access increases. Importantly, the sign of the relationship turns positive among firms with sufficiently strong predetermined data access.[12] This prediction is consistent with the non-monotonic relationship between investment responsiveness to business cycle shocks and data intensity we estimate empirically. Endogenous data accumulation is critical for this result because this pattern does not emerge in the benchmark with exogenous data access.

The relationship between firms' predetermined access to data and their responsiveness to an aggregate productivity shock is shaped by two countervailing forces. On the one hand, an aggregate productivity shock affects the expected productivity of data-rich firms to a relatively small extent (if $\kappa_e > 0$). On the other hand, the presence of data markets amplifies the responsiveness of any firm's capital choices, and particularly so for firms with superior predetermined access to data. The second effect, which makes the relationship between firms' predetermined data access and the responsiveness to aggregate productivity shocks more positive, is particularly strong within a subsample of firms with relatively good data access. This is because differences in data access give rise to particularly large differences in the optimal capital stock (and thus, particularly large differences in the magnitude of the second-round effect) within a group of firms that are relatively data-rich.

There is a complementary explanation for our empirical findings regarding the relationship between firms' data intensity and their responsiveness to an aggregate productivity shock. Specifically, the positive relationship between a firm's data access and investment cyclicality can also emerge due to a positive correlation between firms' lagged data intensity and their efficiency of acquiring data (the parameter η_i), i.e., if firms with a higher lagged data intensity are more efficient in acquiring data. We formalize this in Section C.1 of the Online Appendix, where we prove that $\frac{\partial^2 \phi_d(\cdot)}{\partial \bar{\sigma}_i \partial \eta_i} < 0$.

Active data markets also affect how heterogeneity in the parameter ρ_i, which governs to what extent the uncertainty a firm faces raises its investment costs, translates into differences in size. We formalize this in the following proposition:

Proposition 3. *Firms whose investment costs are more responsive to idiosyncratic uncertainty (i.e., have a higher ρ_i) optimally employ less capital if and only if $\psi(\cdot) < 0$, where*

$$\psi(\cdot) := 0.75(\eta_i)^2 \kappa_e \kappa_v (K^*_{t+1})^{0.5} + \rho_i (\eta_i \kappa_v)^2 K^*_{t+1} - (\bar{V} + \kappa_v \bar{\sigma}_i). \tag{13}$$

Moreover, $\frac{\partial \psi(\cdot)}{\partial \eta_i} > 0$ holds.

[12] This holds because the relationship between predetermined data access and responsiveness to shocks is negative (respectively, positive) if $\frac{\partial \phi_d(\cdot)}{\partial \bar{\sigma}_i}$ is positive (respectively, negative), and $\frac{\partial \phi_d(\cdot)}{\partial \bar{\sigma}_i}$ becomes more negative as $\bar{\sigma}_i$ falls.

If access to data is exogenous, firms whose investment costs are more responsive to the idiosyncratic uncertainty they face hold less capital, given that they face higher capital costs. In the presence of data markets, there is an opposing effect: Acquiring data allows firms to reduce the idiosyncratic uncertainty they face. The gains from uncertainty reduction are larger for firms whose investment costs are more sensitive to uncertainty. Thus, this channel endows such firms with stronger incentives to acquire data, which translate into stronger investment incentives due to the complementarity between data and capital. In particular, the sign of the relationship between a firm's size and the sensitivity of its investment costs to uncertainty becomes positive if it is cheap enough to acquire data (i.e., if η_i is large enough).

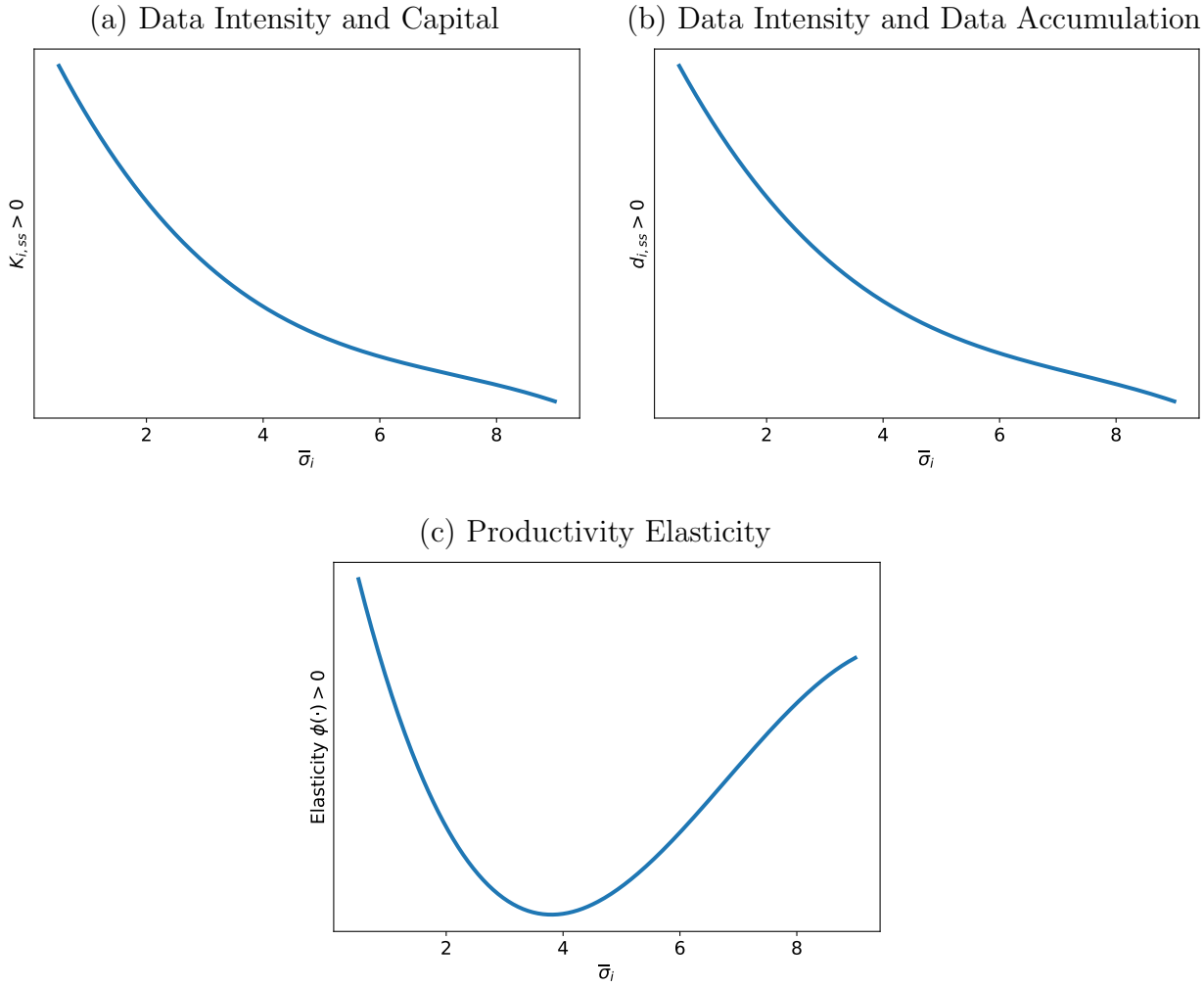
The way in which data markets shape the relationship between firms' sensitivity to uncertainty and their size has non-trivial implications for aggregate stability. In the presence of active data markets, firms whose investment costs are more sensitive to idiosyncratic uncertainty may attain larger capital stocks and, consequently, larger market shares. Thus, aggregate investment becomes increasingly driven by firms whose investment costs are more sensitive to uncertainty, and thus becomes more sensitive to aggregate uncertainty shocks. Given that uncertainty shocks are an important driver of cyclical fluctuations (Bloom, 2009; Bloom et al., 2018), the presence of data markets can thus amplify cyclical fluctuations. Moreover, if heterogeneity in the parameter ρ_i reflects differences in the way in which firms are affected by financial frictions, the presence of data markets may significantly affect how financial stress propagates throughout the economy.

We complete the analysis by presenting the results of numerical analysis which demonstrates that our simplifying assumptions in this section do not qualitatively affect our insights. Specifically, we now solve our model with a depreciation rate $\delta \in (0, 1)$ numerically.[13] Figure 5 shows the key results. In all cases, $\bar{\sigma}_i$ is on the horizontal axis. Panel (a) shows the steady state capital stock and Panel (b) the steady state data acquisition. As established above, capital and data acquisition are complements. Panel (c) shows the elasticity of capital investment with respect to an aggregate productivity shock.

Panel (c) shows that higher predetermined data access has non-monotonic effects on the investment-productivity elasticity. At relatively low levels of data intensity, higher data intensity decreases the elasticity, whereas it increases it at high levels of data intensity.

[13] We use an illustrative calibration with $\beta = 0.96, \alpha = 0.5, r = 0.15, \delta = 0.2, \bar{A} = 1, \bar{V} = 1, \kappa_e = 0.015, \kappa_v = 0.2, \rho = 0.5, \eta = 0.5$.

Figure 5: Direct Acquisition of Data



Panels (a) and (b) of this figure show the steady state capital stock $K_{i,ss}$ and data choice $d_{i,ss}$ as a function of $\bar{\sigma}_i$, respectively. Panel (c) shows the elasticity of firms' capital choice with respect to an aggregate productivity shock for different $\bar{\sigma}_i$. Calibration: $\beta = 0.96, \alpha = 0.5, r = 0.15, \delta = 0.2, \bar{A} = 1, \bar{V} = 1, \kappa_e = 0.015, \kappa_v = 0.2, \rho = 0.5, \eta = 0.5$.

4.4 The Data Feedback Loop

We next show that an active data feedback loop generates amplification effects analogous to those generated by data markets. To facilitate the derivation of analytical results, we focus on a benchmark in which capital depreciates fully across time periods (i.e., set $\delta = 1$) and in which firms cannot acquire data directly (i.e., set $\eta_i = 0$ for all firms). We refer to this benchmark as the *data feedback loop benchmark*.

If there is an active data feedback loop, the relationship between a firm's predetermined access to data and their responsiveness to an aggregate productivity shock is *negative* among firms with relatively weak data access, and *positive* among firms with relatively good data access. We formalize this in the following Proposition:

Proposition 4. *Consider the data feedback loop benchmark.* $\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i} > 0$ *holds if and only if* $\frac{\partial \tilde{\phi}_d(\cdot)}{\partial \bar{\sigma}_i} > 0$*, where*

$$\tilde{\phi}_d(\cdot) := \alpha(\alpha-1)(\bar{A} - \kappa_e \bar{\sigma}_i) + \alpha(\alpha+1)\kappa_e \zeta_i K_{t+1}^* + 2\rho_i \kappa_v \zeta_i (K_{t+1}^*)^{2-\alpha}. \tag{14}$$

Furthermore, $\frac{\partial^2 \phi(\cdot)}{\partial \bar{\sigma}_i^2} > 0$ *and* $\frac{\partial^2 \tilde{\phi}_d(\cdot)}{\partial \bar{\sigma}_i^2} > 0$ *hold.*

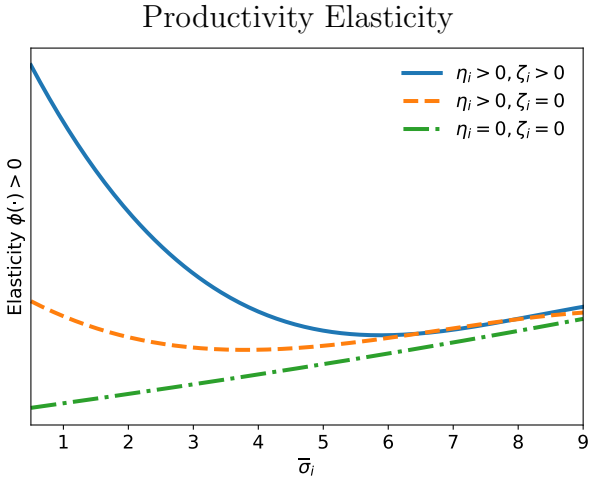
The presence of a data feedback loop amplifies the responsiveness of all firms to an aggregate productivity shock because it triggers a second-round effect: The first-round effect of the shock on a firm's capital input triggers a change in the firm's access to data, which feeds back into the firm's optimal capital choice. The magnitude of the second-round effect is larger for firms with superior predetermined access to data (lower $\bar{\sigma}_i$) because these firms adjust their optimal capital stock more strongly in response to any shock. In addition, any heterogeneity in $\bar{\sigma}_i$ induces larger differences in the magnitude of the second-round effect between firms within a subsample of firms with a relatively low $\bar{\sigma}_i$.

In Section C.2 of the Online Appendix, we prove that the presence of a data feedback loop can also flip the sign of the relationship between a firm's size and the sensitivity of its investment costs to uncertainty. This is because the data feedback loop increases the investment incentives of firms with high ρ_i, given that firms can reduce the idiosyncratic uncertainty they face (through the data feedback loop) by accumulating capital.

We complete this subsection by presenting numerical results that show how the data feedback loop affects firms' responsiveness to shocks, both when there are data markets and when there are not. Figure 6 shows the capital elasticities with respect to productivity shocks, $\phi(\cdot)$, for different levels of $\bar{\sigma}_i$. The green-dashed-dotted line shows the results for the model variant with exogenous data access. More data-intensive firms respond less to productivity shocks in this case. The orange-dashed line shows the results for the model variant with data acquisition but no data feedback loop (as already shown in Figure 5). The blue-solid line shows the results for a model variant in which firms can acquire data directly and there is an active data feedback loop.

Consistent with our empirical and theoretical results, higher predetermined data access (lower $\bar{\sigma}_i$) increases the response to productivity shocks at high levels of data access but decreases it at low levels of data access if there are data markets. The data feedback loop amplifies these patterns by strengthening the relationship between predetermined data access and firms' capital adjustments in response to shocks (holding direct data purchases fixed). In turn, this magnifies differences between the responsiveness of firms' data acquisition choices.

Figure 6: Direct Acquisition of Data and the Data Feedback Loop



This figure shows the elasticity of firms' capital choice with respect to an aggregate productivity shock for different $\bar{\sigma}_i$. The blue-solid line shows the case with data acquisition and an active data feedback loop, the orange-dashed line the case without the data feedback loop, and the green-dashed-dotted line the case with exogenous and time-invariant data, only. We use the same calibration as for Figure 5, and $\zeta = 0.002$ for the case with $\zeta > 0$.

4.5 Extensions

We begin by discussing extensions. In Online Appendix D.1, we consider a model in which firms all face the same costs of capital, but have mean-variance preferences. We analytically establish that the insights we derive within the baseline model extend verbatim. This strengthens the relevance of our predictions, given the substantial empirical evidence that firm managers are risk-averse (Caliendo et al., 2024).

In Appendix D.2, we consider a model in which data accumulates over time rather than depreciating fully within one period. This means that a firm's predetermined access to data becomes an endogenous state variable, and heterogeneity in firms' predetermined access to data at a given point in time may reflect both exogenous factors and past endogenous choices—specifically, the amount of data firms acquired in earlier periods. We use numerical analysis to show that the key results from the baseline model continue to hold.

Importantly, greater persistence in the value of data (i.e., a lower data depreciation rate) amplifies the curvature of the relationship between data access and investment cyclicality. This is because data accumulation further strengthens the magnitude of the second-round effect, and particularly so for firms with superior predetermined data access.

In Appendix D.3, we consider a model in which access to data does not favorably affect a firm's productivity distribution, but only enables firms to predict their future idiosyncratic productivities. We first consider a benchmark in which access to data is exogenously given. As in the main analysis, data-rich firms respond less strongly to aggregate productiv-

ity shocks. Intuitively, this result emerges because any aggregate productivity shock affects the information set of a firm with superior access to data to a lesser extent, thereby eliciting a smaller response. Thereafter, we discuss a version of this model in which firms can endogenously acquire data. As before, a firm's data acquisition choices and its size are strategic complements. This complementarity amplifies the effects of any shock through a second-round effect, and particularly so for data-rich firms.

In Appendix D.4, we consider a model in which data may have decreasing or increasing returns to scale. When data has decreasing returns to scale, an increase in data access affects a firm's expected productivity to a lesser extent if the firm is data-rich—this makes the relationship between data access and the responsiveness to an aggregate productivity shock more convex, which reinforces the corresponding result from the baseline analysis. However, the presence of decreasing returns to scale in data also gives rise to a countervailing effect by reducing the magnitude of the second-round effect, and particularly so for data-rich firms: Any given change in the amount of data that is acquired after a shock (the initial part of the second-round effect) will affect the productivity distribution of data-rich firms to a lesser extent, and thus affect their optimal capital choice less substantially. We provide sufficient conditions under which our theoretical predictions from the main analysis extend.

5 The Role of Data Intensity in General Equilibrium

So far, we focused on firms' investment and data choices in partial equilibrium. We now embed this investment and data block into a general equilibrium setup, allowing us to examine the aggregate consequences of the data-capital complementarity we document in partial equilibrium. We focus on a representative data-using investment firm and we place it in an otherwise standard New Keynesian model with capital investment (following the structure in Winberry et al., 2025, but abstracting from heterogeneity). Details and derivations are relegated to Online Appendix E.

Environment. A representative household supplies labor, holds nominal bonds, and owns all firms. A competitive final-good producer aggregates a continuum of retail varieties with elasticity ε. Monopolistically competitive retailers buy a homogeneous wholesale good and reset prices subject to Rotemberg adjustment costs, which delivers a standard New Keynesian Phillips curve. Monetary policy sets the nominal rate following a standard Taylor rule.

The data-using firm. The key innovation is the upstream producer, who operates the data technology of Section 3. Output uses capital, labor, and data-augmented productivity,

$$Y_t = A_t K_t^{\alpha} N_t^{1-\alpha}, \qquad A_t = \exp\big(a_t + \kappa_e D_t\big), \tag{15}$$

where D_t is the stock of data, with $\sigma_t = \bar{\sigma} - D_t$, and a_t is an aggregate productivity shifter. As in the partial-equilibrium analysis, better data raises expected productivity (governed by κ_e) and lowers the conditional variance of productivity. The latter shows up as a data-dependent cost of investment,

$$\Omega_t = \omega \exp\big(v_t - \kappa_v D_{t+1}\big), \qquad \Omega_D(\cdot) < 0, \tag{16}$$

where v_t is an aggregate uncertainty shifter. Investing I_t units costs $I_t(1+\Omega_t)$. The financing wedge Ω_t is the analogue of the cost-of-capital term $\rho_i \, \kappa_v \, \sigma_i$ in Section 3. Superior data lowers conditional uncertainty and thereby lowers the effective cost of installing capital, with κ_v controlling the strength of this channel.

The producer accumulates capital K_t and data D_t. Data evolves through persistence, direct acquisition, and the data feedback loop:

$$D_{t+1} = (1 - \delta_D)\, D_t + \eta \, d_t + \zeta \, K_{t+1}, \tag{17}$$

where $\delta_D \in [0, 1]$ is the depreciation rate of data, d_t is directly acquired data (purchased at a convex cost), η governs the efficiency of direct acquisition, and ζK_{t+1} is the data feedback loop. These are the three data sources of the analytical model in Section 3, but expressed as an accumulable stock rather than a static signal noise. Setting $\kappa_e = \kappa_v = \eta = \zeta = 0$ recovers a standard New Keynesian model with capital.

Connection to the analytical model. The GE block nests the mechanisms of Section 4. The strategic complementarity between data and capital, established in Lemma 1, reappears in the producer's optimality conditions (see Appendix E). The capital Euler equation contains an extra term reflecting the data feedback loop: a marginal unit of capital raises future data and hence is worth its data shadow value, while the data Euler equation contains a term $\kappa_v \, \Omega_t I_t$: a marginal unit of data lowers the uncertainty wedge on current investment. The first channel is the analogue to "more capital raises the marginal value of data," the second one to "better data raises the marginal return to capital." Their interaction generates the second-round amplification that drives our analytical results: an aggregate shock that moves capital also moves data, which feeds back into capital. In the general equilibrium

model, this complementarity operates at the level of the aggregate producer and is mediated by equilibrium movements in the real rate, the wage, and the relative price of the wholesale good.

Solution. To account for the nonlinearities arising due to the data-capital complementarity, we solve the model using a third-order perturbation around the deterministic steady state, which preserves the dependence of investment on (time-varying) uncertainty and on the data stock that mitigates it.

Calibration. We calibrate the non-data-related parameters to standard values. In particular, we set the discount factor $\beta = 0.99$, the capital share $\alpha = 0.33$, the capital depreciation rate to $\delta_K = 0.025$, risk aversion and the inverse Frisch elasticity to 1, and the final good producer's demand elasticity to $\varepsilon = 6$. We set the persistence of the aggregate productivity and uncertainty shocks to 0.9, and the interest-rate smoothing parameter in the Taylor rule to 0.8. The inflation coefficient in the Taylor rule is set to 1.5. The Rotemberg adjustment cost parameter is set to $\varphi_p = 50$. The investment adjustment cost parameter is set such that investment in the non-data economy decreases by 1pp. in response to a monetary policy shock of 25bp (not shown), as in Winberry et al. (2025). This results in $\varphi_I = 12.317$.

This leaves us with six data-related parameters. We set $\omega = 0.02$ to normalize the cost of investment in the no-data steady state to $\Omega = 0.02$. Given a target steady-state data stock of $D = 10$, we then set κ_v such that the steady-state cost of investment in the data economy is 70% of that in the no-data economy, i.e., $\Omega = \omega \exp(-\kappa_v D) = 0.014$, which yields $\kappa_v = 0.036$. We set η and ζ jointly such that the steady-state inflows into the data stock are split evenly between directly acquired data, ηd, and the data feedback loop, ζK, with each channel accounting for half of the steady-state stock $D = 10$. This results in $\zeta = 0.01$ and $\eta = 1.58$. We view this choice as instructive to examine whether the two channels matter differently for business cycle fluctuations while accounting for equal shares of the data stock in the steady state. Given these values, the steady-state data law of motion, $\delta_D D = \eta d + \zeta K$, implies $\delta_D = 0.109$. We assume that aggregate productivity in the steady state is 30% higher in the data economy compared to the no-data economy, resulting in $\kappa_e = 0.026$. We set the data adjustment cost parameter such that consumption in the data economy increases by 1 percent in response to an aggregate productivity shock of 0.005. This results in $\varphi_d = 4.645$. We also consider uncertainty shocks of $\varepsilon_t^v = 0.005$.

5.1 Results

Aggregate productivity shocks. Figure 7 reports the responses to a positive aggregate productivity shock ε^a. Output (panel a), investment (panel b), and consumption (panel c) exhibit a hump-shaped response. Data amplifies the transmission of the shock. On impact, output rises by about 2.5% in our baseline model (blue-solid lines) against roughly 1.1% in the no-data economy (purple-dotted lines), and investment rises by about 1.2% against roughly 0.4%, a threefold difference.

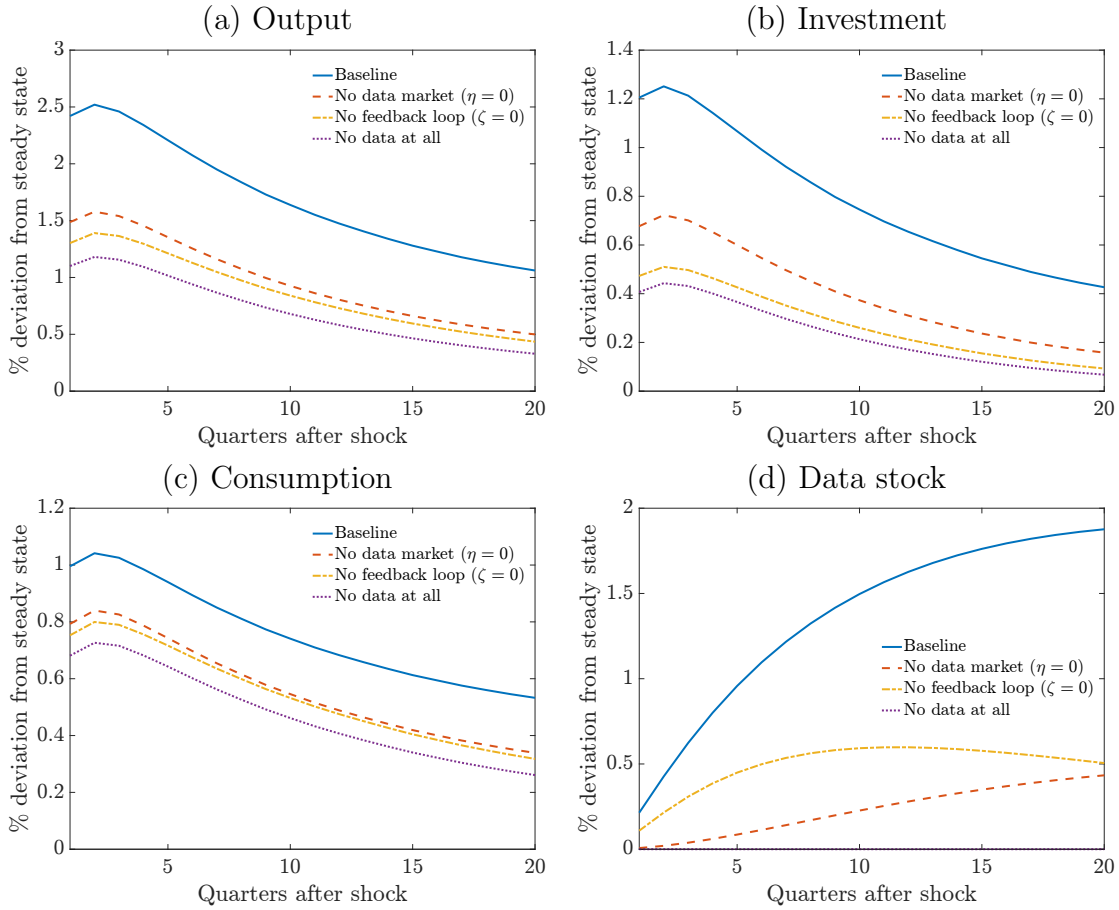
Panel (d) makes the propagation mechanism explicit. A positive productivity shock raises investment and hence, capital. Through the feedback loop as well as direct data acquisition, the stock of data accumulates gradually and persistently (shown in panel (d)). Better data in turn raises expected productivity (through κ_e) and lowers the investment cost Ω_t (through κ_v), feeding back into capital. This endogenous build-up of the data stock is the aggregate general-equilibrium second-round effect that drives our analytical results in Section 4.

The two intermediate calibrations decompose this amplification further. Shutting down direct data acquisition ($\eta = 0$, red-dashed lines) and shutting down the data feedback loop ($\zeta = 0$, yellow-dashed-dotted lines) both pull the responses well below the baseline, with the no-data-feedback economy lying below the no-data-market economy: the feedback loop is the more powerful amplifier of a productivity shock, while direct acquisition adds a further increment on top of it. Since in steady state both forces account for 50% of the data stock, the relative importance of the feedback loop reflects its dynamic advantage: it scales with capital, which itself expands in response to the shock. Crucially, the two channels are strongly complementary. The baseline response exceeds the no-data response substantially more than the sum of the increments contributed by each channel in isolation, because each channel raises the marginal value of the other. Hence, the data feedback loop and direct data acquisition reinforce each other and ignoring one would understate the effects of the remaining one.

Aggregate uncertainty shocks. Figure 8 reports the responses to a positive aggregate uncertainty shock ε^v, which raises v_t and hence the investment cost $\Omega_t = \omega \exp(v_t - \kappa_v D_{t+1})$. The shock is contractionary: investment and output fall on impact and recover slowly.

Output and investment decline on impact and stay persistently low. The baseline economy exhibits the largest and most persistent decline in output and investment, while the no-feedback economy ($\zeta = 0$) is nearly indistinguishable from the no-data economy. Hence, the feedback loop is again the dominant channel: without it, the data market alone barely alters the economy's response to an uncertainty shock. Consumption rises slightly on impact before turning negative as the contraction propagates. This reflects intertemporal consump-

Figure 7: Impulse Responses to Aggregate Productivity Shocks

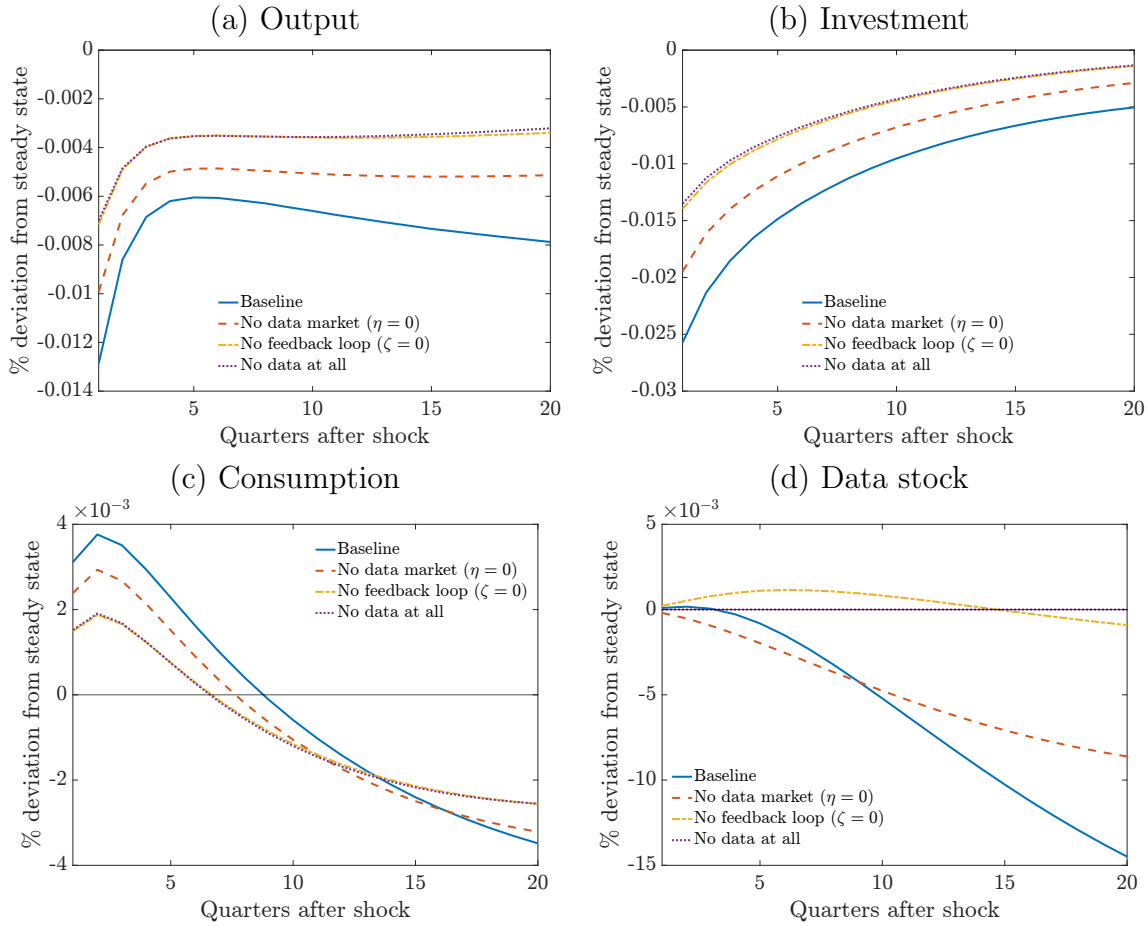


This figure shows impulse responses to a positive aggregate productivity shock (ε^a). Panel (a) shows output Y, (b) investment I, (c) consumption C, and (d) the data stock D—in percent deviations from the respective model's deterministic steady state. The four lines correspond to alternative calibrations of the data block: the baseline full model (solid blue); a model without a data market ($\eta = 0$, dashed red); a model without a data feedback loop ($\zeta = 0$, dash-dotted yellow); and a standard New Keynesian model without data ($\kappa_e = \kappa_v = \eta = \zeta = 0$, dotted purple).

tion smoothing: the uncertainty shock raises the cost of new investment but does not destroy existing resources, so households initially redirect goods away from investment toward consumption. As the lower capital stock depresses future output and wages, consumption subsequently falls.

The data stock in panel (d) reveals a tension between the two channels. In the baseline, data decumulates after an uncertainty shock: the collapse in investment lowers capital and, through the feedback loop, drags the data stock down. When only direct acquisition is active ($\zeta = 0$), by contrast, the data stock rises modestly, because higher uncertainty raises the marginal value of data — data lowers the investment cost Ω_t — and firms respond by acquiring more of it. This stabilizing acquisition motive is dominated in the full model,

Figure 8: Impulse Responses to Aggregate Uncertainty Shocks



This figure shows impulse responses to a positive aggregate uncertainty shock (ε^v). Panel (a) shows output Y, (b) investment I, (c) consumption C, and (d) the data stock D—in percent deviations from the respective model's deterministic steady state. The four lines correspond to alternative calibrations of the data block: the baseline full model (solid blue); a model without a data market ($\eta = 0$, dashed red); a model without a data feedback loop ($\zeta = 0$, dash-dotted yellow); and a standard New Keynesian model without data ($\kappa_e = \kappa_v = \eta = \zeta = 0$, dotted purple).

where the feedback loop ties the data stock to a sharply falling capital stock. Taken together, these results illustrate that the data-capital complementarity amplifies the transmission of *both* productivity and uncertainty shocks in general equilibrium, with the feedback loop as the dominant channel and direct acquisition reinforcing it.

6 The Effects of Digital Markets Regulation

Digital markets regulations—such as the EU General Data Protection Regulation (GDPR), the Digital Markets Act (DMA), the California Consumer Privacy Act (CCPA), and the EU Data Act—govern how firms collect, process, and share data. Perhaps surprisingly, the

policy debate surrounding them has almost entirely abstracted from their macroeconomic consequences at business cycle frequency. Our analysis provides a natural lens to analyze these instruments: different regulatory measures map directly into changes in the parameters governing firms' data accumulation, and the general equilibrium model of Section 5 allows us to trace their aggregate effects.

We organize the discussion around three types of regulatory measures and their mapping into the model. The first is regulation that limits firms' ability to acquire data directly, such as the data minimization principle in Article 5 of the GDPR and Section 1798.100(c) of the California Civil Code. Within the model, this corresponds to a reduction in the efficiency of direct data acquisition, η. The second is regulation that mandates dominant firms to share data with competitors, such as Article 6 of the DMA and the data-sharing requirements of the EU Data Act. These measures can be viewed as exogenous improvements of data access for firms with low and intermediate data intensities. The third is regulation that limits or prohibits the internal reuse of data across business lines—such as the restriction the European Commission imposed on Google as part of its merger with Fitbit (European Commission, 2020)—which within our framework maps into a reduction in the strength of the data feedback loop, ζ.

Restrictions on direct data acquisition ($\downarrow \eta$). Provisions that limit how much data firms can collect reduce η, lowering the efficiency with which firms translate expenditure into data stock. In the GE model, a lower η weakens the direct-acquisition channel of data accumulation. As shown in Section 5.1, the direct-acquisition channel amplifies the transmission of both productivity and uncertainty shocks by enabling the data stock to respond endogenously to capital: when $\eta = 0$, the impulse responses of output and investment lie above the no-data benchmark but below our baseline model. Regulation that suppresses η therefore attenuates this amplification, dampening the propagation of aggregate shocks. At the same time, it reduces the endogenous build-up of the data stock that sustains the complementarity between data and capital, so the dampening is likely persistent.

Mandated data sharing. Mandates which require dominant firms to share data with competitors have countervailing effects: increasing the data access of data-poor firms weakens their responsiveness to business cycle shocks, whereas increasing the data access of firms with intermediate access to data amplifies their responsiveness. If data sharing mandates primarily benefit firms with intermediate access to data, they will thus amplify aggregate fluctuations, and dampen them if they primarily benefit firms with little or none.

Restrictions on the data feedback loop ($\downarrow \zeta$). Regulation that prohibits firms from reusing data collected in one activity to improve decisions in another—for instance, preventing a platform from using advertising data to improve product recommendations—limits the extent to which operational scale translates into data accumulation. This maps into a reduction in ζ, the coefficient governing how capital feeds into next period's data stock. The GE results in Section 5.1 show that the feedback loop is the dominant amplification channel: shutting it down ($\zeta = 0$) brings the impulse responses of output and investment close to the no-data benchmark, far more so than eliminating direct acquisition alone. Regulation that reduces ζ therefore has the largest dampening effect on aggregate fluctuations of the three measures considered. Importantly, this dampening is symmetric across shock types: both the productivity-shock and uncertainty-shock responses are substantially attenuated when the feedback loop is weakened, as the endogenous co-movement between capital and data that underlies the second-round amplification is severed.

Regulatory trade-offs. Taken together, these examples reveal a macroeconomic trade-off in digital markets regulation. Measures that suppress data accumulation—whether by reducing η or ζ—dampen the amplification of aggregate shocks and reduce the volatility of output and investment. At the same time, they weaken the data–capital complementarity that makes the economy more productive in the long run. Further, measures that expand data access may be desirable from a competition policy perspective, but amplify cyclical fluctuations if they primarily benefit firms with intermediate data intensities.

7 Conclusion

Modern economies increasingly revolve around big data, i.e., digitized information about consumers and market conditions. In this paper, we study how firms' use of big data shapes the propagation of cyclical fluctuations. Using matched employer–employee data to measure firms' data intensity, we document that the responsiveness of firms' investment to business cycle shocks is U-shaped in their data intensity.

We rationalize this pattern in a model of firm investment with endogenous data acquisition, in which data and capital are strategic complements. Embedded in general equilibrium, this complementarity materially amplifies the transmission of aggregate shocks, with the data feedback loop as the dominant channel. Our framework also offers a macroeconomic perspective on digital markets regulation, such as the EU GDPR, and reveals a trade-off: measures that suppress data accumulation dampen cyclical fluctuations but also weaken the data-capital complementarity that raises productivity.

As economic activity becomes increasingly digital and advances in AI expand firms' ability to extract value from data, the macroeconomic importance of big data is likely to grow. Our analysis points to several promising avenues for future research. Embedding data–capital complementarities into environments with adjustment frictions would clarify the extent to which endogenous data acquisition generates state-dependent investment responses. Linking data acquisition to financial frictions—screening, monitoring, endogenous spreads—could determine whether tight credit curtails data investment and amplifies cyclical fluctuations. More broadly, integrating endogenous data acquisition into medium-scale macroeconomic models may help explain how data shapes labor demand, innovation, productivity, and market structure.

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ONLINE APPENDIX

A Appendix to Section 2

Section 2.4 shows that the U-shape survives interacting the shock with lagged size, leverage, cash, age, and intangibility. Table A.1 reports the full ladder of control specifications underlying that result, summarizing the two objects that define the U-shape, namely, the dip and the rebound. Adding the firm controls (row b) and then balance-sheet levels (row c) barely moves the estimates relative to no controls (row a). Row (d) is the transmission-channel specification featured in the main text: the shock interacted with size, leverage, cash, age, and intangibility. The results remain robust throughout.

Table A.1: Control Robustness: The U-Shape Under Alternative Control Sets

	Absolute		Relative	
	Y shock	I shock	Y shock	I shock
Dip (pooled): $\overline{Q1\text{–}Q4} - (\text{DI}=0)$				
(a) No controls (FE only)	-0.041***	-0.043***	-0.042***	-0.045***
	(0.013)	(0.015)	(0.014)	(0.015)
(b) Baseline: L.emp, L.IT-share	-0.042***	-0.046***	-0.044***	-0.049***
	(0.015)	(0.016)	(0.016)	(0.016)
(c) + balance-sheet levels	-0.034**	-0.037**	-0.036**	-0.040***
	(0.014)	(0.015)	(0.015)	(0.015)
(d) + characteristics × shock	-0.023*	-0.028*	-0.024*	-0.029*
	(0.013)	(0.014)	(0.014)	(0.015)
Rebound: $Q5 - Q1$				
(a) No controls (FE only)	0.044**	0.046***	0.044**	0.045**
	(0.017)	(0.018)	(0.020)	(0.020)
(b) Baseline: L.emp, L.IT-share	0.044***	0.045***	0.044**	0.043**
	(0.017)	(0.017)	(0.020)	(0.020)
(c) + balance-sheet levels	0.037**	0.038**	0.036*	0.038**
	(0.016)	(0.016)	(0.019)	(0.019)
(d) + characteristics × shock	0.036**	0.036**	0.039**	0.041**
	(0.016)	(0.017)	(0.020)	(0.020)
Firm & industry×year FE	Yes	Yes	No	No
Firm & industry×year-quarter FE	No	No	Yes	Yes

Dip (pooled) is the average of $Q1$–$Q4$ minus the DI=0 bin, and *Rebound* is $Q5 - Q1$, for within-year quintiles of *Data intensity*$_{f,t-1}$ (sample 1993–2017). Rows add controls cumulatively: (a) firm and industry-by-(year/year-quarter) fixed effects only; (b) adds lagged ln(employment) and the lagged IT employment share; (c) adds lagged balance-sheet levels (leverage, the cash-to-assets ratio, and the intangible-asset ratio); (d) adds the shock interacted with five lagged characteristics (ln(employment), leverage, the cash-to-assets ratio, firm age, and the intangible-asset ratio). As in the main table, "Absolute" columns use firm and industry-by-year fixed effects and report cyclicality levels, and "Relative" columns use firm and industry-by-year-quarter fixed effects and report differences relative to the DI=0 bin. Standard errors, two-way clustered by firm and year-quarter, are in parentheses.

Additional robustness. Table A.2 collects further checks.

Sales. If data-intensive firms were simply more exposed to cyclical demand, their sales should display the same U-shape as their investment. Panel A of Table A.2 shows that this is not the case. Sales are procyclical in all bins, with coefficients between 0.02 and 0.04 for the Y shock, but cyclicality is flat across bins: the dip and the rebound are small, insignificant, and close to zero in all four columns.

Firm size and LinkedIn coverage. Firms with few LinkedIn profiles can only have either zero or a high measured data intensity. To make sure that such firms do not drive our results, Panel B drops firms with fewer than 20 LinkedIn profiles (22% of the sample), and Panel C drops firms with fewer than 20 employees in Compustat. The dip remains significant at the 1% level in all columns; the rebound is significant in all columns in Panel C and in three of four columns in Panel B (the relative specification for the Y shock has a p-value of about 0.10). Panel D shows that the results are unchanged when we compute data intensity without Revelio's sampling weights.

Since more than half of the firm-quarters have no measured data workers, one may worry that the DI=0 bin mainly captures firms that are poorly covered on LinkedIn. Panel E splits this bin at the within-year median of LinkedIn coverage, defined as the LinkedIn headcount relative to Compustat employment. DI=0 firms with low coverage are, if anything, slightly less cyclical than those with high coverage (the difference is insignificant), and the dip relative to high-coverage DI=0 firms is at least as large as in the baseline.

Table A.2: Investment Cyclicality and Data Intensity—Additional Robustness

	Absolute		Relative	
	Y shock	I shock	Y shock	I shock
Panel A: ln(Sales) as dependent variable				
Dip (pooled): $\overline{Q1–Q4}$−DI=0	−0.009	−0.006	−0.009	−0.006
	(0.007)	(0.007)	(0.007)	(0.008)
Rebound: $Q5 - Q1$	−0.004	−0.003	0.001	0.003
	(0.010)	(0.010)	(0.011)	(0.012)
Observations	195,256	195,256	189,145	189,145
Panel B: firms with at least 20 LinkedIn profiles				
Dip (pooled): $\overline{Q1–Q4}$−DI=0	−0.034***	−0.040***	−0.037***	−0.044***
	(0.013)	(0.015)	(0.014)	(0.015)
Rebound: $Q5 - Q1$	0.038**	0.039**	0.034	0.036*
	(0.016)	(0.017)	(0.020)	(0.020)
Observations	155,714	155,714	149,628	149,628
Panel C: firms with at least 20 employees in Compustat				
Dip (pooled): $\overline{Q1–Q4}$−DI=0	−0.040***	−0.044***	−0.041***	−0.046***
	(0.014)	(0.015)	(0.015)	(0.015)
Rebound: $Q5 - Q1$	0.050***	0.052***	0.049**	0.051***
	(0.017)	(0.018)	(0.020)	(0.020)
Observations	184,579	184,579	178,631	178,631
Panel D: data intensity without sampling weights				
Dip (pooled): $\overline{Q1–Q4}$−DI=0	−0.042***	−0.045***	−0.043***	−0.048***
	(0.014)	(0.016)	(0.016)	(0.016)
Rebound: $Q5 - Q1$	0.042**	0.041**	0.040**	0.038*
	(0.017)	(0.018)	(0.020)	(0.021)
Observations	199,901	199,901	193,809	193,809
Panel E: DI=0 bin split by LinkedIn coverage				
DI=0 low coverage − DI=0 high coverage	−0.016	−0.021	−0.010	−0.017
	(0.015)	(0.016)	(0.016)	(0.017)
Dip (pooled): $\overline{Q1–Q4}$−DI=0 high coverage	−0.046**	−0.051***	−0.043**	−0.051***
	(0.018)	(0.019)	(0.019)	(0.020)
Rebound: $Q5 - Q1$	0.050***	0.051***	0.050**	0.050**
	(0.018)	(0.018)	(0.020)	(0.020)
Observations	189,552	189,552	183,626	183,626

Notes. Panels A–E: the sample is 1993–2017 (firm-quarters). Specifications as in Table 4: firms with no data workers form the DI=0 bin, the remaining firms are sorted into within-year quintiles $Q1, \ldots, Q5$ of data intensity, and each bin indicator is interacted with the shock. "Dip (pooled)" is the average of $Q1$–$Q4$ minus DI=0, and "Rebound" is $Q5 - Q1$. All specifications include firm fixed effects, lagged ln(employment), and the lagged IT employment share. *Absolute* columns add industry-by-year fixed effects; *relative* columns add industry-by-year-quarter fixed effects. Panel A replaces ln(capital expenditure) by ln(sales), quarterly revenue, on the baseline sample. Panels B and C restrict the sample to firms with at least 20 (unweighted) LinkedIn profiles in the year in which data intensity is measured, and to firms with at least 20 employees in Compustat, respectively. Panel D measures data intensity as the unweighted share of data-related profiles. Panel E splits the DI=0 bin at the within-year median of LinkedIn coverage (LinkedIn headcount relative to Compustat employment); the high-coverage DI=0 bin is the omitted category in the relative columns. Standard errors, two-way clustered by firm and year-quarter, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B Proofs

Proof of Proposition 1: Throughout the following analysis, we consider a one-time (transitory) aggregate productivity shock. Formally, we set $\bar{A}_{t+1} = \bar{A}$, $\bar{A}_j = \bar{A}'$ for all $j > t+1$, and consider a change of $\bar{A}$. We set $\bar{V}_{t+1} = \bar{V}$. By Assumption 1, $\bar{A} - \kappa_e \bar{\sigma}_i > 0$ holds.

If $\zeta_i = 0$ and $\eta_i = 0$, the Bellman equation reads:

$$V(K_{i,t}) = \max_{K_{i,t+1}} \left\{ \big(\bar{A} - \kappa_e \bar{\sigma}_i\big)(K_{i,t+1})^{\alpha} - \big(r + \rho_i \bar{V} + \rho_i \kappa_v \bar{\sigma}_i\big)\big(K_{i,t+1} - (1-\delta)K_{i,t}\big) + \beta V^f(K_{i,t+1}) \right\},$$

where $\frac{\partial V^f(K_{i,t+1})}{\partial K_{i,t+1}} = (1-\delta)(r + \rho_i \bar{V} + \rho_i \kappa_v \bar{\sigma}_i)$.

The optimal capital stock, which we label $K^*_{t+1}(\cdot)$, thus needs to solve the following first-order condition:

$$\alpha\big(\bar{A} - \kappa_e \bar{\sigma}_i\big)(K^*_{t+1}(\cdot))^{\alpha-1} - \big(r + \rho_i \bar{V} + \rho_i \kappa_v \bar{\sigma}_i\big) + \beta(1-\delta)(r + \rho_i \bar{V} + \rho_i \kappa_v \bar{\sigma}_i) = 0$$

The closed-form solution for the optimal capital stock reads:

$$K^*_{t+1} = \left(\frac{\alpha\big(\bar{A} - \kappa_e \bar{\sigma}_i\big)}{r - \beta(1-\delta)r + \big(1 - \beta(1-\delta)\big)\rho_i\big(\bar{V} + \kappa_v \bar{\sigma}_i\big)} \right)^{\frac{1}{1-\alpha}}$$

The relative effect of an aggregate productivity shock is given by:

$$\phi(\cdot) = \frac{\partial K^*_{t+1}}{\partial \bar{A}} \frac{\bar{A}}{K^*_{t+1}} = \frac{1}{1-\alpha} \frac{\bar{A}}{\bar{A} - \kappa_e \bar{\sigma}_i} > 0 \tag{B.1}$$

Note that $\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i} \geq 0$ holds, with a strict inequality if $\kappa_e > 0$, because:

$$\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i} = \frac{\kappa_e}{1-\alpha} \frac{\bar{A}}{[\bar{A} - \kappa_e \bar{\sigma}_i]^2} \geq 0$$

■

Proof of Lemma 1: Since $\zeta_i = 0$, $\eta_i > 0$, and $\delta = 1$, any firm's problem is static: In any period t, a firm i chooses $d_{i,t+1}$ and $K_{i,t+1}$ to maximize the objective function

$$\Pi_{t+1} = \big(\bar{A} - \kappa_e(\bar{\sigma}_i - \eta_i d_{i,t+1})\big)(K_{i,t+1})^{\alpha} - \big(r + \rho_i \bar{V} + \rho_i \kappa_v (\bar{\sigma}_i - \eta_i d_{i,t+1})\big) K_{i,t+1} - (d_{i,t+1})^2. \tag{B.2}$$

Taking the first-order condition with respect to $d_{i,t+1}$ and solving yields:

$$d^*_{t+1}(\cdot) = 0.5\eta_i\big(\kappa_e K^{\alpha}_{i,t+1} + \rho_i\kappa_v K_{i,t+1}\big). \tag{B.3}$$

Taking the first-order condition with respect to $K_{i,t+1}$ implies that the optimal capital stock must solve the following first-order condition:

$$\alpha\big(\bar{A} - \kappa_e(\bar{\sigma}_i - \eta_i d^*_{t+1}(\cdot))\big)\,(K^*_{t+1})^{\alpha-1} - \big(r + \rho_i\bar{V} + \rho_i\kappa_v(\bar{\sigma}_i - \eta_i d^*_{t+1}(\cdot))\big) = 0 \tag{B.4}$$

■

Proof of Proposition 2:

Part 1: Deriving the relative effect of an aggregate productivity shock. Note that $d^*_{t+1}(K_{t+1}) = 0.5\eta_i\big(\kappa_e K^{0.5}_{t+1} + \rho_i\kappa_v K_{t+1}\big)$. Plugging this into the capital optimality condition given in equation (B.4) yields the following equation which must hold in optimum:

$$T := 0.5\big(\bar{A} - \kappa_e\bar{\sigma}_i\big)(K^*_{t+1})^{-0.5} + 0.5\kappa_e\eta_i\Big[0.5\kappa_e\eta_i(K^*_{t+1})^{0.5} + 0.5\eta_i\rho_i\kappa_v(K^*_{t+1})\Big](K^*_{t+1})^{-0.5}$$

$$-\big(r + \rho_i\bar{V} + \rho_i\kappa_v\bar{\sigma}_i\big) + \rho_i\kappa_v\eta_i\Big[0.5\kappa_e\eta_i(K^*_{t+1})^{0.5} + 0.5\eta_i\rho_i\kappa_v(K^*_{t+1})\Big] = 0$$

Thus, we have:

$$\frac{\partial T}{\partial \bar{A}} = 0.5(K^*_{t+1})^{-0.5} \quad ; \quad \frac{\partial T}{\partial \bar{\sigma}_i} = -0.5\kappa_e(K^*_{t+1})^{-0.5} - \rho_i\kappa_v$$

$$\frac{\partial T}{\partial K_{t+1}} = -\frac{1}{4}\big(\bar{A} - \kappa_e\bar{\sigma}_i\big)(K^*_{t+1})^{-1.5} + \frac{3}{8}(\eta_i)^2\kappa_e\kappa_v\rho_i(K^*_{t+1})^{-0.5} + \frac{1}{2}(\eta_i\kappa_v\rho_i)^2$$

By Assumption 1, $\frac{\partial T}{\partial K_{t+1}} < 0$ holds. By the previous results, the relative effect of an aggregate productivity shock is:

$$\phi(\cdot) = -\frac{0.5\bar{A}(K^*_{t+1})^{-1.5}}{-\frac{1}{4}\big(\bar{A} - \kappa_e\bar{\sigma}_i\big)(K^*_{t+1})^{-1.5} + \frac{3}{8}(\eta_i)^2\kappa_e\kappa_v\rho_i(K^*_{t+1})^{-0.5} + \frac{1}{2}(\eta_i\kappa_v\rho_i)^2} \iff$$

$$\phi(\cdot) = -\frac{\frac{1}{2}\bar{A}}{-\frac{1}{4}\big(\bar{A} - \kappa_e\bar{\sigma}_i\big) + \frac{3}{8}(\eta_i)^2\kappa_e\kappa_v\rho_i(K^*_{t+1}) + \frac{1}{2}(\eta_i\kappa_v\rho_i)^2(K^*_{t+1})^{1.5}} \tag{B.5}$$

Define $\phi_d(\cdot) := -\frac{1}{4}\big(\bar{A} - \kappa_e\bar{\sigma}_i\big) + \frac{3}{8}(\eta_i)^2\kappa_e\kappa_v\rho_i(K^*_{t+1}) + \frac{1}{2}(\eta_i\kappa_v\rho_i)^2(K^*_{t+1})^{1.5}$. Note that $\phi_d(\cdot) = \frac{\partial T}{\partial K_{t+1}}(K_{t+1})^{1.5}$, which implies that $\phi_d(\cdot) < 0$ by Assumption 1. Note further that $\phi(\cdot) = -\frac{0.5\bar{A}}{\phi_d(\cdot)}$. If $\phi_d(\cdot)$ increases, so does $\phi(\cdot)$.

Part 2: Bounding $\frac{\partial K^*_{t+1}}{\partial\bar{\sigma}_i}$ from above. We can write:

$$\frac{\partial T}{\partial K_{t+1}}(K_{t+1})^{0.5} = -\frac{1}{4}\big(\bar{A} - \kappa_e\bar{\sigma}_i\big)(K^*_{t+1})^{-1} + \frac{3}{8}(\eta_i)^2\kappa_e\kappa_v\rho_i + \frac{1}{2}(\eta_i\kappa_v\rho_i)^2(K^*_{t+1})^{0.5} >$$

$$-\frac{1}{4}\big(\bar{A} - \kappa_e\bar{\sigma}_i\big)(K^*_{t+1})^{-1}$$

Given $\bar{A} - \kappa_e\bar{\sigma}_i > 0$ holds by Assumption 1, this implies that

$$\frac{\partial K^*_{t+1}}{\partial\bar{\sigma}_i} < \frac{0.5\kappa_e + \rho_i\kappa_v(K^*_{t+1})^{0.5}}{-\frac{1}{4}\big(\bar{A} - \kappa_e\bar{\sigma}_i\big)(K^*_{t+1})^{-1}} < \frac{0.5\kappa_e}{-\frac{1}{4}\big(\bar{A} - \kappa_e\bar{\sigma}_i\big)(K^*_{t+1})^{-1}} = \frac{2\kappa_e}{-\big(\bar{A} - \kappa_e\bar{\sigma}_i\big)(K^*_{t+1})^{-1}}$$

Part 3: Establishing that $\frac{\partial^2 K^*_{t+1}}{\partial\bar{\sigma}_i^2} > 0$ holds. Note that

$$\frac{\partial K^*_{t+1}}{\partial\bar{\sigma}_i} = \frac{0.5\kappa_e + \rho_i\kappa_v(K^*_{t+1})^{0.5}}{-\frac{1}{4}\big(\bar{A} - \kappa_e\bar{\sigma}_i\big)(K^*_{t+1})^{-1} + \frac{3}{8}(\eta_i)^2\kappa_e\kappa_v\rho_i + \frac{1}{2}(\eta_i\kappa_v\rho_i)^2(K^*_{t+1})^{0.5}},$$

Note further that the denominator of $\frac{\partial K^*_{t+1}}{\partial\bar{\sigma}_i}$ is equal to $\frac{\partial T}{\partial K^*_{t+1}}(K^*_{t+1})^{0.5}$ and thus negative by Assumption 1. Thus, we have:

$$\frac{\partial^2 K^*_{t+1}}{\partial\bar{\sigma}_i^2} = \frac{1}{[-\frac{1}{4}\big(\bar{A} - \kappa_e\bar{\sigma}_i\big)(K^*_{t+1})^{-1} + \frac{3}{8}(\eta_i)^2\kappa_e\kappa_v\rho_i + \frac{1}{2}(\eta_i\kappa_v\rho_i)^2(K^*_{t+1})^{0.5}]^2}$$

$$\Bigg[\bigg(-\frac{1}{4}\big(\bar{A}-\kappa_e\bar{\sigma}_i\big)(K^*_{t+1})^{-1}+\frac{3}{8}(\eta_i)^2\kappa_e\kappa_v\rho_i+\frac{1}{2}(\eta_i\kappa_v\rho_i)^2(K^*_{t+1})^{0.5}\bigg)\bigg(0.5\rho_i\kappa_v(K^*_{t+1})^{-0.5}\frac{\partial K^*_{t+1}}{\partial\bar{\sigma}_i}\bigg)-$$

$$\Big(0.5\kappa_e + \rho_i\kappa_v(K^*_{t+1})^{0.5}\Big)$$

$$\bigg(\frac{1}{4}\kappa_e(K^*_{t+1})^{-1} + \frac{1}{4}\big(\bar{A} - \kappa_e\bar{\sigma}_i\big)(K^*_{t+1})^{-2}\frac{\partial K^*_{t+1}}{\partial\bar{\sigma}_i} + \frac{1}{4}(\eta_i\kappa_v\rho_i)^2(K^*_{t+1})^{-0.5}\frac{\partial K^*_{t+1}}{\partial\bar{\sigma}_i}\bigg)\Bigg] \tag{B.6}$$

In the following, we show that $\frac{1}{4}\kappa_e(K^*_{t+1})^{-1} + \frac{1}{4}\big(\bar{A} - \kappa_e\bar{\sigma}_i\big)(K^*_{t+1})^{-2}\frac{\partial K^*_{t+1}}{\partial\bar{\sigma}_i} < 0$. To see why this holds true, note that:

$$\frac{1}{4}\kappa_e(K^*_{t+1})^{-1} + \frac{1}{4}\big(\bar{A} - \kappa_e\bar{\sigma}_i\big)(K^*_{t+1})^{-2}\frac{\partial K^*_{t+1}}{\partial\bar{\sigma}_i} < 0 \iff \kappa_e + \big(\bar{A} - \kappa_e\bar{\sigma}_i\big)(K^*_{t+1})^{-1}\frac{\partial K^*_{t+1}}{\partial\bar{\sigma}_i} < 0$$

In turn, this implies that:

$$\kappa_e + \big(\bar{A} - \kappa_e\bar{\sigma}_i\big)(K^*_{t+1})^{-1}\frac{\partial K^*_{t+1}}{\partial\bar{\sigma}_i} < \kappa_e + \big(\bar{A} - \kappa_e\bar{\sigma}_i\big)(K^*_{t+1})^{-1}\frac{2\kappa_e}{-\big(\bar{A} - \kappa_e\bar{\sigma}_i\big)(K^*_{t+1})^{-1}} < 0$$

This implies that $\frac{\partial^2 K^*_{t+1}}{\partial\bar{\sigma}_i^2} > 0$, given that all terms in equation (B.6) are positive by the previous arguments.

Part 4: Establishing that $\frac{\partial^2\phi_d}{\partial\bar{\sigma}_i^2} > 0$ holds. Note that

$$\frac{\partial\phi_d(\cdot)}{\partial\bar{\sigma}_i} = \frac{1}{4}\kappa_e + \frac{3}{8}(\eta_i)^2\kappa_e\kappa_v\rho_i\frac{\partial K^*_{t+1}}{\partial\bar{\sigma}_i} + \frac{3}{4}(\eta_i\kappa_v\rho_i)^2(K^*_{t+1})^{0.5}\frac{\partial K^*_{t+1}}{\partial\bar{\sigma}_i}$$

This implies that:

$$\frac{\partial^2\phi_d(\cdot)}{\partial\bar{\sigma}_i^2} = \frac{3}{8}(\eta_i)^2\kappa_e\kappa_v\rho_i\frac{\partial^2 K^*_{t+1}}{\partial\bar{\sigma}_i^2} + \frac{3}{8}(\eta_i\kappa_v\rho_i)^2(K^*_{t+1})^{-0.5}\left(\frac{\partial K^*_{t+1}}{\partial\bar{\sigma}_i}\right)^2 + \frac{3}{4}(\eta_i\kappa_v\rho_i)^2(K^*_{t+1})^{0.5}\frac{\partial^2 K^*_{t+1}}{\partial\bar{\sigma}_i^2}$$

Given that $\frac{\partial^2 K^*_{t+1}}{\partial\bar{\sigma}_i^2} > 0$, it follows that $\frac{\partial^2\phi_d(\cdot)}{\partial\bar{\sigma}_i^2} > 0$.

Part 5: Establishing that $\frac{\partial^2\phi(\cdot)}{\partial\bar{\sigma}_i^2} > 0$ holds. We wish to show that $\frac{\partial\phi(\cdot)}{\partial\bar{\sigma}_i} > 0$. Note firstly that

$$\frac{\partial\phi(\cdot)}{\partial\bar{\sigma}_i} = \frac{0.5\bar{A}}{[\phi_d(\cdot)]^2}\frac{\partial\phi_d(\cdot)}{\partial\bar{\sigma}_i}$$

Note further that $\phi_d(\cdot) < 0$ holds by definition and that $\frac{\partial^2\phi_d(\cdot)}{\partial\bar{\sigma}_i^2} > 0$. This implies that:

$$\frac{\partial^2\phi(\cdot)}{\partial\bar{\sigma}_i^2} = -\frac{\bar{A}}{[\phi_d(\cdot)]^3}\left(\frac{\partial\phi_d(\cdot)}{\partial\bar{\sigma}_i}\right)^2 + \frac{0.5\bar{A}}{[\phi_d(\cdot)]^2}\frac{\partial^2\phi_d(\cdot)}{\partial\bar{\sigma}_i^2} > 0$$

■

Proof of Proposition 3:

Note that:

$$\frac{\partial T}{\partial\rho_i} = 0.75(\eta_i)^2\kappa_e\kappa_v(K^*_{t+1})^{0.5} + \rho_i(\eta_i\kappa_v)^2K^*_{t+1} - \big(\bar{V} + \kappa_v\bar{\sigma}_i\big)$$

This implies that:

$$\frac{\partial K^*_{t+1}}{\partial\rho_i} = -\frac{0.75(\eta_i)^2\kappa_e\kappa_v(K^*_{t+1})^{0.5} + \rho_i(\eta_i\kappa_v)^2K^*_{t+1} - (\bar{V} + \kappa_v\bar{\sigma}_i)}{-\frac{1}{4}(\bar{A} - \kappa_e\bar{\sigma}_i)(K^*_{t+1})^{-1.5} + \frac{3}{8}(\eta_i)^2\kappa_e\kappa_v\rho_i(K^*_{t+1})^{-0.5} + \frac{1}{2}(\eta_i\kappa_v\rho_i)^2}$$

This implies that $\frac{\partial K^*_{t+1}}{\partial \rho_i} < 0$ if and only if:

$$\underbrace{0.75(\eta_i)^2\kappa_e\kappa_v(K^*_{t+1})^{0.5} + \rho_i(\eta_i\kappa_v)^2K^*_{t+1} - (\bar{V} + \kappa_v\bar{\sigma}_i)}_{:=\psi(\cdot)} < 0$$

Note further that:

$$\frac{\partial \psi(\cdot)}{\partial \eta_i} = \frac{3}{2}(\eta_i)\kappa_e\kappa_v(K^*_{t+1})^{0.5} + 2\rho_i\eta_i(\kappa_v)^2K^*_{t+1} + \left(\frac{3}{8}(\eta_i)^2\kappa_e\kappa_v(K^*_{t+1})^{-0.5} + \rho_i(\eta_i\kappa_v)^2\right)\frac{\partial K^*_{t+1}}{\partial \eta_i} > 0,$$

given that $\frac{\partial K^*_{t+1}}{\partial \eta_i} > 0$. ■

Proof of Proposition 4:

Part 1: Obtaining expressions for the relative effects of shocks. We again consider one-time aggregate productivity shocks defined as before.

We set $\delta = 1$, which renders the problem fully static. We further impose that $\eta_i = 0$, but suppose that $\zeta_i > 0$ (which means that the data feedback loop is active). The expected profits the firm obtains in period $t+1$ are thus given by.

$$\Pi_{i,t+1}(K_{i,t+1}) = \big(\bar{A} - \kappa_e(\bar{\sigma}_i - \zeta_i K_{i,t+1})\big)(K_{i,t+1})^\alpha - \big(r_t + \rho_i\bar{V} + \rho_i\kappa_v(\bar{\sigma}_i - \zeta_i K_{i,t+1})\big)K_{i,t+1}$$

The optimal capital stock $K^*_{t+1}(\cdot)$ must solve the following first-order condition:

$$\alpha(\bar{A} - \kappa_e\bar{\sigma}_i)(K^*_{t+1})^{\alpha-1} + (\alpha+1)\kappa_e\zeta_i(K^*_{t+1})^\alpha - \big(r_t + \rho_i\bar{V} + \rho_i\kappa_v\bar{\sigma}_i\big) + 2\rho_i\kappa_v\zeta_i K^*_{t+1} = 0 \tag{B.7}$$

The relative effect of an aggregate productivity shock is given by:

$$\phi(\cdot) = -\frac{\alpha\,\bar{A}}{\alpha(\alpha-1)(\bar{A} - \kappa_e\bar{\sigma}_i) + \alpha(\alpha+1)\kappa_e\zeta_i K^*_{t+1} + 2\rho_i\kappa_v\zeta_i(K^*_{t+1})^{2-\alpha}}$$

We define $\tilde{\phi}_d(\cdot) := \alpha(\alpha-1)(\bar{A} - \kappa_e\bar{\sigma}_i) + \alpha(\alpha+1)\kappa_e\zeta_i K^*_{t+1} + 2\rho_i\kappa_v\zeta_i(K^*_{t+1})^{2-\alpha}$. Note that $\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i} > 0$ holds true if and only if $\frac{\partial \tilde{\phi}_d(\cdot)}{\partial \bar{\sigma}_i} > 0$.

Part 2: The functions $x(\bar{\sigma}_i) := \alpha(\alpha-1)(\bar{A} - \kappa_e\bar{\sigma}_i)(K^*_{t+1})^{\alpha-1}$ and $y(\bar{\sigma}_i) := \alpha(\alpha-1)(\bar{A} - \kappa_e\bar{\sigma}_i)(K^*_{t+1})^{-1}$ are decreasing in $\bar{\sigma}_i$. To begin, we bound $\frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i}$ from above. Specifically,

we show that $\frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} < \frac{\kappa_e(K^*_{t+1})}{(\alpha-1)(\bar{A}-\kappa_e\bar{\sigma}_i)}$. To see this, note that:

$$\frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} = \frac{\alpha\kappa_e(K^*_{t+1})^{\alpha-1} + \rho_i\kappa_v}{\alpha(\alpha-1)(\bar{A}-\kappa_e\bar{\sigma}_i)(K^*_{t+1})^{\alpha-2} + \alpha(\alpha+1)\kappa_e\zeta_i(K^*_{t+1})^{\alpha-1} + 2\rho_i\kappa_v\zeta_i}$$

where the denominator is $\frac{\partial T}{\partial K^*_{t+1}} < 0$. Note that:

$$\alpha(\alpha-1)(\bar{A}-\kappa_e\bar{\sigma}_i)(K^*_{t+1})^{\alpha-2}+\alpha(\alpha+1)\kappa_e\zeta_i(K^*_{t+1})^{\alpha-1}+2\rho_i\kappa_v\zeta_i > \alpha(\alpha-1)(\bar{A}-\kappa_e\bar{\sigma}_i)(K^*_{t+1})^{\alpha-2}$$

Given that both sides of this inequality are strictly negative, we can rewrite this inequality as follows:

$$\frac{1}{\alpha(\alpha-1)(\bar{A}-\kappa_e\bar{\sigma}_i)(K^*_{t+1})^{\alpha-2} + \alpha(\alpha+1)\kappa_e\zeta_i(K^*_{t+1})^{\alpha-1} + 2\rho_i\kappa_v\zeta_i} < \frac{1}{\alpha(\alpha-1)(\bar{A}-\kappa_e\bar{\sigma}_i)(K^*_{t+1})^{\alpha-2}}$$

Thus, we have:

$$\frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} < \frac{\alpha\kappa_e(K^*_{t+1})^{\alpha-1} + \rho_i\kappa_v}{\alpha(\alpha-1)(\bar{A}-\kappa_e\bar{\sigma}_i)(K^*_{t+1})^{\alpha-2}} < \frac{\alpha\kappa_e(K^*_{t+1})^{\alpha-1}}{\alpha(\alpha-1)(\bar{A}-\kappa_e\bar{\sigma}_i)(K^*_{t+1})^{\alpha-2}} = \frac{\kappa_e(K^*_{t+1})}{(\alpha-1)(\bar{A}-\kappa_e\bar{\sigma}_i)}$$

Having established this, note that

$$\frac{\partial x(\bar{\sigma}_i)}{\partial \bar{\sigma}_i} = \alpha(\alpha-1)\left[-\kappa_e\left(K^*_{t+1}\right)^{\alpha-1} + \left(\bar{A}-\kappa_e\bar{\sigma}_i\right)(\alpha-1)\left(K^*_{t+1}\right)^{\alpha-2}\frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i}\right]$$

One can show that this is negative, given that $\alpha(\alpha-1) < 0$ and because

$$-\kappa_e\left(K^*_{t+1}\right)^{\alpha-1} + \left(\bar{A}-\kappa_e\bar{\sigma}_i\right)(\alpha-1)\left(K^*_{t+1}\right)^{\alpha-2}\frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} > 0 \iff$$

$$\left(\bar{A}-\kappa_e\bar{\sigma}_i\right)(\alpha-1)\left(K^*_{t+1}\right)^{-1}\frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} > \kappa_e$$

The last inequality holds because

$$\left(\bar{A}-\kappa_e\bar{\sigma}_i\right)(\alpha-1)\left(K^*_{t+1}\right)^{-1}\frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} > \underbrace{\left(\bar{A}-\kappa_e\bar{\sigma}_i\right)(\alpha-1)\left(K^*_{t+1}\right)^{-1}\left[\frac{\kappa_e(K^*_{t+1})}{(\alpha-1)(\bar{A}-\kappa_e\bar{\sigma}_i)}\right]}_{=\kappa_e}$$

Now we consider the function $y(\bar{\sigma}_i) := \alpha(\alpha-1)(\bar{A}-\kappa_e\bar{\sigma}_i)(K^*_{t+1})^{-1}$. The derivative of this function is strictly negative, given that:

$$-\alpha(\alpha-1)(\bar{A}-\kappa_e\bar{\sigma}_i)(K^*_{t+1})^{-2}\frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} - \alpha(\alpha-1)\kappa_e(K^*_{t+1})^{-1} < 0 \iff$$

$$(\bar{A} - \kappa_e \bar{\sigma}_i)(K^*_{t+1})^{-1} \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} + \kappa_e < 0$$

To see that this inequality holds, note that:

$$(\bar{A} - \kappa_e \bar{\sigma}_i)(K^*_{t+1})^{-1} \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} + \kappa_e < (\bar{A} - \kappa_e \bar{\sigma}_i)(K^*_{t+1})^{-1} \frac{\kappa_e (K^*_{t+1})}{(\alpha - 1)(\bar{A} - \kappa_e \bar{\sigma}_i)} + \kappa_e = \frac{\kappa_e \alpha}{\alpha - 1} < 0$$

Part 3: $\frac{\partial^2 \tilde{\phi}_d(\cdot)}{\partial \bar{\sigma}_i^2} > 0$ **holds.** Note that:

$$\frac{\partial \tilde{\phi}_d(\cdot)}{\partial \bar{\sigma}_i} = -\alpha(\alpha - 1)\kappa_e + \alpha(\alpha + 1)\kappa_e \zeta_i \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} + 2(2 - \alpha)\rho_i \kappa_v \zeta_i (K^*_{t+1})^{1-\alpha} \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i}$$

This implies that $\frac{\partial^2 \tilde{\phi}_d(\cdot)}{\partial \bar{\sigma}_i^2} > 0$. This is because $\frac{\partial^2 \tilde{\phi}_d(\cdot)}{\partial \bar{\sigma}_i^2} > 0$, given that:

$$\frac{\partial^2 \tilde{\phi}_d(\cdot)}{\partial \bar{\sigma}_i^2} = \alpha(\alpha + 1)\kappa_e \zeta_i \frac{\partial^2 K^*_{t+1}}{\partial \bar{\sigma}_i{}^2} + 2\rho_i \kappa_v \zeta_i \Bigg[(2 - \alpha)(1 - \alpha)(K^*_{t+1})^{-\alpha} \left(\frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} \right)^2 +$$

$$(2 - \alpha)(K^*_{t+1})^{1-\alpha} \frac{\partial^2 K^*_{t+1}}{\partial \bar{\sigma}_i{}^2} \Bigg] > 0$$

This is strictly positive because all terms are strictly positive. To see this, note that:

$$\frac{\partial^2 K^*_{t+1}}{\partial \bar{\sigma}_i^2} = \frac{1}{[\alpha(\alpha - 1)(\bar{A} - \kappa_e \bar{\sigma}_i)(K^*_{t+1})^{-1} + \alpha(\alpha + 1)\kappa_e \zeta_i + 2\rho_i \kappa_v \zeta_i (K^*_{t+1})^{1-\alpha}]^2}$$

$$\Bigg[\left(\alpha(\alpha-1)(\bar{A}-\kappa_e \bar{\sigma}_i)(K^*_{t+1})^{-1} + \alpha(\alpha+1)\kappa_e \zeta_i + 2\rho_i \kappa_v \zeta_i (K^*_{t+1})^{1-\alpha} \right) \left(\rho_i \kappa_v (1-\alpha)(K^*_{t+1})^{-\alpha} \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} \right)$$

$$- \left(\alpha \kappa_e + \rho_i \kappa_v (K^*_{t+1})^{1-\alpha} \right)$$

$$\left(-\alpha(\alpha-1)(\bar{A}-\kappa_e \bar{\sigma}_i)(K^*_{t+1})^{-2} \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} - \alpha(\alpha-1)\kappa_e (K^*_{t+1})^{-1} + 2\rho_i \kappa_v \zeta_i (1-\alpha)(K^*_{t+1})^{-\alpha} \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} \right) \Bigg]$$

All terms in this expression are strictly positive by the previous arguments.

Part 4: $\frac{\partial^2 \phi(\cdot)}{\partial \bar{\sigma}_i^2} > 0$ **holds.** Note that $\phi(\cdot) = -\frac{\alpha \bar{A}}{\tilde{\phi}_d(\cdot)}$. This implies that:

$$\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i} = \frac{\alpha \bar{A}}{[\tilde{\phi}_d(\cdot)]^2} \frac{\partial \tilde{\phi}_d(\cdot)}{\partial \bar{\sigma}_i}$$

In turn, this implies that:

$$\frac{\partial^2 \phi(\cdot)}{\partial \bar{\sigma}_i^2} = -\frac{2\alpha \bar{A}}{[\tilde{\phi}_d(\cdot)]^3}\left(\frac{\partial \tilde{\phi}_d(\cdot)}{\partial \bar{\sigma}_i}\right)^2 + \frac{\alpha\, \bar{A}}{[\tilde{\phi}_d(\cdot)]^2}\frac{\partial^2 \tilde{\phi}_d(\cdot)}{\partial \bar{\sigma}_i^2}$$

This is strictly positive since $\tilde{\phi}_d(\cdot) < 0$ and because $\frac{\partial^2 \tilde{\phi}_d(\cdot)}{\partial \bar{\sigma}_i^2} > 0$ by part 3. ■

C Auxiliary Theoretical Results

C.1 Comparative Statics: η_i

We now show that our empirical predictions can also emerge in our theoretical model if there is a positive correlation between a firm's data intensity and its efficiency in acquiring more data (the parameter η_i in our model), i.e., if firms with a higher data intensity are more efficient in acquiring a given amount of data:

Proposition 5. *Consider the data markets benchmark and suppose that $\alpha = 0.5$, $\kappa_e > 0$, and $\kappa_v > 0$. Then, $\frac{\partial^2 \phi_d(\cdot)}{\partial \bar{\sigma}_i \partial \eta_i} < 0$.*

Verbally, this result states that the relationship between a firm's predetermined access to data and its responsiveness to an aggregate productivity shock is *negative* when considering firms for which η_i is sufficiently small and *positive* when considering firms for which η_i is sufficiently large. Intuitively, this result holds because increases in η_i amplify the differences between the magnitude of the second-round effect across firms with heterogeneous $\bar{\sigma}_i$.

C.2 Risk Exposure and the Data Feedback Loop

We now show that the presence of the data feedback loop can also flip the sign of the relationship between ρ_i and a firm's size:

Proposition 6. *Consider the data feedback loop benchmark. The property $\frac{\partial K^*_{t+1}}{\partial \rho_i} > 0$ holds if and only if:*

$$-\left(\bar{V} + \kappa_v \bar{\sigma}_i\right) + 2\kappa_v \zeta_i K^*_{t+1} > 0. \tag{C.1}$$

Note that $2\kappa_v \zeta_i K^*_{t+1}$ is increasing in ζ_i. This implies that the sign of the derivative $\frac{\partial K^*_{t+1}}{\partial \rho_i}$ is positive when the data feedback loop is strong enough. The intuition which underlies this result is familiar: If there is an active data feedback loop, firms can reduce the uncertainty

they face by attaining size. The benefits of this mechanism are particularly strong for firms with a high ρ_i, which strengthens their incentives to invest.

C.3 Proofs

Proof of Proposition 5:

Recall that:

$$\phi_d(\cdot) = -\frac{1}{4}\left(\bar{A} - \kappa_e\bar{\sigma}_i\right) + \frac{3}{8}(\eta_i)^2\kappa_e\kappa_v\rho_i(K^*_{t+1}) + \frac{1}{2}(\eta_i\kappa_v\rho_i)^2(K^*_{t+1})^{1.5} < 0 \tag{C.2}$$

Note that:

$$\frac{\partial^2\phi_d}{\partial\bar{\sigma}_i\partial\eta_i} = \frac{6}{8}(\eta_i)\kappa_e\kappa_v\rho_i\frac{\partial K^*_{t+1}}{\partial\bar{\sigma}_i} + \frac{3}{8}(\eta_i)^2\kappa_e\kappa_v\rho_i\frac{\partial^2 K^*_{t+1}}{\partial\bar{\sigma}_i\partial\eta_i} +$$

$$\frac{3}{2}\eta_i(\kappa_v\rho_i)^2(K^*_{t+1})^{0.5}\frac{\partial K^*_{t+1}}{\partial\bar{\sigma}_i} + \frac{3}{8}(\eta_i\kappa_v\rho_i)^2(K^*_{t+1})^{-0.5}\frac{\partial K^*_{t+1}}{\partial\eta_i}\frac{\partial K^*_{t+1}}{\partial\bar{\sigma}_i} + \frac{3}{4}(\eta_i\kappa_v\rho_i)^2(K^*_{t+1})^{0.5}\frac{\partial^2 K^*_{t+1}}{\partial\bar{\sigma}_i\partial\eta_i}$$

All terms are strictly negative. To see why, note that $\frac{\partial K^*_{t+1}}{\partial\bar{\sigma}_i} < 0$, $\frac{\partial K^*_{t+1}}{\partial\eta_i} > 0$, and $\frac{\partial^2 K^*_{t+1}}{\partial\bar{\sigma}_i\partial\eta_i} < 0$. To see why the last feature holds, note that $\frac{\partial T}{\partial K^*_{t+1}} < 0$ holds by Assumption 1 and that:

$$\frac{\partial K^*_{t+1}}{\partial\bar{\sigma}_i} = \frac{0.5\kappa_e + \rho_i\kappa_v(K^*_{t+1})^{0.5}}{-\frac{1}{4}\left(\bar{A} - \kappa_e\bar{\sigma}_i\right)(K^*_{t+1})^{-1} + \frac{3}{8}(\eta_i)^2\kappa_e\kappa_v\rho_i + \frac{1}{2}(\eta_i\kappa_v\rho_i)^2(K^*_{t+1})^{0.5}}$$

Thus, we have:

$$\frac{\partial^2 K^*_{t+1}}{\partial\bar{\sigma}_i\partial\eta_i} = \frac{1}{[(K^*_{t+1})^{0.5}(\partial T/\partial K^*_{t+1})]^2}\left[(K^*_{t+1})^{0.5}\frac{\partial T}{\partial K^*_{t+1}}\left[0.5\rho_i\kappa_v(K^*_{t+1})^{-0.5}\frac{\partial K^*_{t+1}}{\partial\eta_i}\right] -\right.$$

$$\left(0.5\kappa_e + \rho_i\kappa_v(K^*_{t+1})^{0.5}\right)$$

$$\left.\left(\frac{1}{4}\left(\bar{A} - \kappa_e\bar{\sigma}_i\right)(K^*_{t+1})^{-2}\frac{\partial K^*_{t+1}}{\partial\eta_i} + \frac{6}{8}(\eta_i)\kappa_e\kappa_v\rho_i + \eta_i(\kappa_v\rho_i)^2(K^*_{t+1})^{0.5} + \frac{1}{4}(\eta_i\kappa_v\rho_i)^2(K^*_{t+1})^{-0.5}\frac{\partial K^*_{t+1}}{\partial\eta_i}\right)\right]$$

All terms in this expression are strictly negative, which proves the desired result. ■

Proof of Proposition 6 : We are interested in the sign of $\frac{\partial K^*_{t+1}}{\partial\rho_i}$. The sign of this derivative is positive if and only if:

$$-\left(\bar{V} + \kappa_v\bar{\sigma}_i\right) + 2\kappa_v\zeta_i K^*_{t+1} > 0$$

This follows directly by inspection of the first-order condition that pins down K^*_{t+1}. ■

D Extensions

D.1 Mean-variance Preferences

In this section, we show that our key insights extend if all firms face the same cost of capital, but have mean-variance preferences as in Eeckhout and Veldkamp (2022). Thus, the two modeling approaches deliver qualitatively analogous predictions.

Model. Formally, we now consider a version of our model in which any firm's cost of capital in a period t is equal to the interest rate r_t. Output in a given period t is given by $A_{i,t}(K_{i,t})^{\alpha}$. Firms have mean-variance preferences as in Eeckhout and Veldkamp (2022): The expected flow utility of any firm i in a period $j > t$ (as evaluated in the time period t) is given by

$$\mathbb{E}_t[Y_{i,j}] - \frac{\lambda_i}{2} VAR_t[Y_{i,j}] - r_{j-1}(K_j - (1-\delta)K_{j-1}), \tag{D.1}$$

where

$$VAR_t[Y_{i,j}] = VAR_t[A_{i,j}](K_{i,j})^{2\alpha}. \tag{D.2}$$

For simplicity, we impose that $\eta_i = 0$ throughout the analysis, but allow for the presence of an active data feedback loop (i.e., that $\zeta_i > 0$). Qualitatively, this is without loss of generality, given that the possibility of acquiring data affects firms' responsiveness to aggregate shocks in the same way as the presence of a data feedback loop. We impose the equivalent of Assumption 1—expected flow profits are strictly concave in capital, and the first two moments of the productivity distribution are always strictly positive. Everything else is as in our baseline model. We refer to this model as the *model with mean-variance preferences*.

Analysis. We establish that heterogeneity in firms' predetermined access to data affects their responsiveness to aggregate shocks as in the baseline analysis. We set $\alpha = 1/2$ to facilitate the analytical derivations—however, our results do not qualitatively rely on this assumption.[14]

Proposition 7. *Consider the model with mean-variance preferences, and set $\alpha = 1/2$. Then:*

- *If $\zeta_i = 0$, firms with superior access to data respond less strongly to aggregate productivity shocks, i.e., $\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i} \geq 0$ holds (with a strict inequality if $\kappa_e > 0$).*

[14] We provide further details to underpin this claim in an earlier working paper version of this manuscript.

- *If $\kappa_e = 0$ and $\zeta_i > 0$, firms with superior access to data respond more strongly to aggregate productivity shocks, i.e., $\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i} < 0$ holds. Moreover, $\frac{\partial^2 \phi(\cdot)}{\partial \bar{\sigma}_i^2} > 0$ holds.*

Suppose that $\zeta_i = 0$ and $\alpha = 1/2$. Then, the firm's optimal capital stock K_{t+1}^* must then solve the following first-order condition.

$$\alpha \mathbb{E}_t[A_{i,t+1}](K_{i,t+1}^*)^{\alpha-1} - \frac{\lambda_i}{2} VAR_t[A_{i,t+1}] - r_t = 0. \tag{D.3}$$

If there is no data feedback loop, but access to better data increases a firm's expected productivity, data-rich firms respond less strongly to aggregate productivity shocks—this is because the shock affects the expected productivity of these firms to a smaller extent (in relative terms). The presence of a data feedback loop amplifies the responsiveness of all firms, and particularly so for data-rich firms. Moreover, there is a positive relationship between a firm's predetermined access to data and the magnitude of the second-round effect of a shock. Together, these channels make the relationship between firms' predetermined data access and their responsiveness to an aggregate productivity shock negative for data-poor firms and less negative (and potentially positive) for data-rich firms.

D.2 Non-depreciating Data

We now consider a model in which data does not fully depreciate within a single period. Instead, data generated in one period continues to provide value in subsequent periods. In such an environment, heterogeneity in firms' predetermined access to data at a given point in time may reflect both exogenous factors and past endogenous choices—specifically, the amount of data firms acquired in earlier periods. We use numerical analysis to show that the key results from the baseline model continue to hold. In particular, the relationship between data access and investment cyclicality is convex and can be non-monotonic. Importantly, greater persistence in the value of data (i.e., lower data depreciation) strengthens these effects quantitatively and amplifies the curvature of the relationship between data access and investment cyclicality.

Model. We consider a model that is identical to the model we laid out in Section 3, with two exceptions: Firstly, we assume that firms cannot acquire data directly, while the data feedback loop may be active. This specification enhances the tractability of the model and is in line with our previous insights that the possibility of direct data acquisition and the presence of a data feedback loop affect firms' responsiveness to aggregate shocks in qualitatively analogous ways. Secondly, the quality of data to which a firm has access in a

given period $t+1$, which we refer to as $\sigma_{i,t+1}$, now depends on the firm's past choices through the state transition function

$$\sigma_{i,t+1} = \bar{\sigma} + (1-\tau)\sigma_{i,t} - \zeta_i K_{i,t+1}, \tag{D.4}$$

where $\tau \in (0,1)$. When a firm chooses its capital stock $K_{i,t+1}$ in period t, heterogeneity in its predetermined access to data is thus captured by differences in $\sigma_{i,t}$. We continue to impose Assumption 1.

Analysis. When data accumulates over time, a firm's predetermined access to data becomes a state variable. As a result, the firm's value function depends on two interacting state variables—capital and predetermined data access. This makes a full analytical characterization of the model intractable. We therefore focus on numerical analysis to study how persistence in data accumulation affects firms' responsiveness to aggregate shocks.

A firm's Bellman equation is given by:

$$\mathcal{V}(K_{i,t}, \sigma_{i,t}) = \max_{K_{i,t+1}} \left\{ f_t(K_{i,t}, \sigma_{i,t}, K_{i,t+1}) + \beta \mathcal{V}'(K_{i,t+1}, \sigma_{i,t+1}) \right\},$$

where

$$f_t(K_{i,t}, \sigma_{i,t}, K_{i,t+1}) = \big(\bar{A} - \kappa_e(\bar{\sigma} + (1-\tau)\sigma_{i,t} - \zeta_i K_{i,t+1})\big)(K_{i,t+1})^{\alpha} -$$

$$\big(r_t + \rho_i \bar{V} + \rho_i \kappa_v (\bar{\sigma} + (1-\tau)\sigma_{i,t} - \zeta_i K_{i,t+1})\big)\big(K_{i,t+1} - (1-\delta)K_{i,t}\big). \tag{D.5}$$

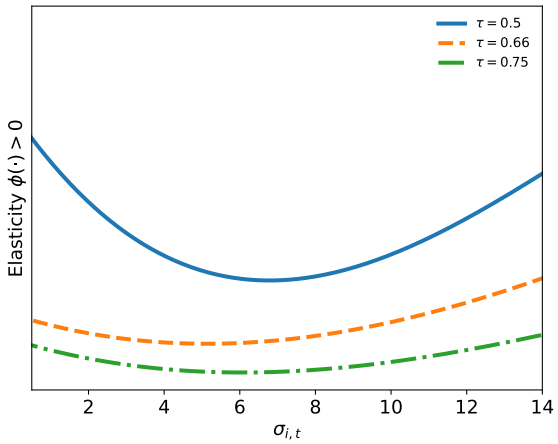
Data persistence implies that firms' current capital choices affect their future data access through the state transition function. In particular, a higher capital stock in period $t+1$ improves the firm's data access in period $t+2$, thereby increasing the marginal return to capital in that period. In turn, an increase in $K_{i,t+2}$ gives rise to stronger incentives to acquire data in period $t+1$ (through the data feedback loop) by increasing $K_{i,t+1}$.

We solve this problem numerically using value function iteration on a two-dimensional grid with capital $(K_{i,t})$ and predetermined data access $(\sigma_{i,t})$.[15] Figure D.1 presents the elasticity of K^*_{t+1} with respect to an aggregate productivity shock, $\phi(\cdot)$. We fix $\bar{\sigma}$ and vary the endogenous state variable $\sigma_{i,t}$ along the horizontal axis, while keeping $K_{i,t}$ fixed at its steady state level.[16] This mirrors our empirical strategy, where aggregate shocks are interacted with predetermined data intensity.

[15] The calibration is as follows: $\beta = 0.96, \alpha = 0.5, \delta = 0.2, \bar{A} = 1, \bar{V} = 1, r = 0.15, \tau = 0.5, \zeta = 0.01, \kappa_e = 0.01, \kappa_v = 0.05, \rho = 0.5$.

[16] The capital stock $K_{i,t+1}$ adjusts in response to the shocks and how it adjusts is affected by $\sigma_{i,t}$.

Figure D.1: Data Accumulation and Aggregate Productivity Shock Elasticity



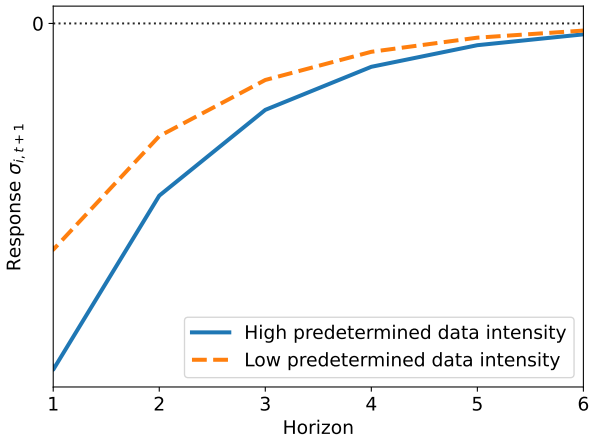
This figure shows the elasticity $\phi(\cdot)$ for different τ as a function of $\sigma_{i,t}$.

The results in Figure D.1 confirm the main predictions of the baseline model and are consistent with our empirical findings: The relationship between a firm's predetermined data access and its responsiveness to an aggregate productivity shock is negative for low levels of data access and becomes positive at higher levels of data access. Importantly, greater persistence in the value of data (i.e., a lower data depreciation rate τ) strengthens the effects we find quantitatively and amplifies the curvature of the relationship between data access and investment cyclicality.

The intuition why greater persistence in the value of data strengthens the relationships we study is as follows: First, greater persistence amplifies the second-round effect for all firms: When the value of data persists over time, an improvement in data access in period $t+1$ increases the marginal return to capital in periods that are further in the future. In turn, the higher capital levels chosen in future periods strengthen firms' incentives to acquire capital in period $t+1$ because this improves data access in the future. Second, this amplification mechanism is particularly strong for firms with superior predetermined access to data. This is because a firm with better predetermined data access in a given period t optimally chooses a higher $K_{i,t+1}$, so aggregate shocks induce larger adjustments in $K_{i,t+1}$. As a result, shocks generate larger changes in the data access of firms with superior predetermined data access at the time of the shock. Consequently, increases in data persistence amplify the strength of the second round effect especially strongly for data-rich firms.

In Figure D.2, we visualize the impulse responses of data intensity, which is inversely related to $\sigma_{i,t+1}$, in response to an expansionary aggregate productivity shock. The blue-solid line shows the responses of a firm with a relatively high predetermined data intensity (low $\sigma_{i,t}$), and the orange-dashed line shows the responses of a firm with a relatively low

Figure D.2: Impulse Response Functions: Data intensities



This figure shows the impulse responses of data intensity, $\sigma_{i,t+1}$, in response to an expansionary aggregate productivity shock. The blue-solid line shows the responses of a firm with a relatively high predetermined data intensity, and the orange-dashed line shows the responses of a firm with a relatively low predetermined data intensity. The initial capital stock for both firms is assumed to be at its deterministic steady state value (which is identical across the two firms).

predetermined data intensity (high $\sigma_{i,t}$). The initial capital stock for both firms is assumed to be at its deterministic steady state value (which is identical across the two firms). Hence, the differences in the data-intensity responses are solely driven by heterogeneous levels of predetermined data intensities. We see that firms with higher predetermined data intensity respond more strongly, i.e., they accumulate more data in response to expansionary shocks.

D.3 Purely Predictive Data

In this section, we consider a model in which access to superior data does not favorably affect the distribution of a firm's productivity, but only enables the firm to predict the realization of its future productivity. If access to data is exogenously given, firms with superior access to data respond less strongly to aggregate productivity shocks. Moreover, we outline an extension with endogenous data acquisition and argue that this possibility reinforces the responsiveness to aggregate shocks, and particularly so for firms with better exogenous access to data.

Model. There are two time periods $t \in \{1, 2\}$ and a continuum of firms i that produce output in period 2. In period 1, any firm chooses how much capital to utilize in period 2. The total profits of a firm that utilizes $K_{i,2}$ units of capital are given by:

$$A_{i,2}(K_{i,2})^{\alpha} - r_i K_{i,2}, \tag{D.6}$$

where $A_{i,2}$ is the productivity of the firm. We specify that $A_{i,2} = \bar{A} + \epsilon_i$, where $\bar{A}$ is public information and ϵ_i is drawn from a continuous distribution with cumulative distribution function G and support $[\underline{\epsilon}, \bar{\epsilon}]$. Moreover, r_i are the costs of capital a firm i faces.

A share $\omega \in [0, 1]$ of all firms has access to data, and the rest does not. A firm without data only discovers the realization of $A_{i,2}$ in period 2. Thus, a firm without data does not know the realization of its productivity when deciding how much capital to utilize. By contrast, a firm with data knows the realization of $A_{i,2}$ when choosing its capital input. The costs of capital are given by $r_d = r + \tilde{\rho}_d$ for a firm with data and by $r_{nd} = r + \tilde{\rho}_{nd}$ for a firm without data, where r is the common part of the interest rate. We refer to this model as the *model with purely predictive data*.

Analysis. We define K_2^{nd} as the optimal capital stock of a firm without data. This solves:

$$K_2^{nd} = \arg\max_{K_2} \left[\int_{\underline{\epsilon}}^{\bar{\epsilon}} (\bar{A} + \epsilon_i)(K_2)^{\alpha} dG(\epsilon_i) - r_{nd} K_2 \right] \tag{D.7}$$

We define $K_2^d(A_{i,2})$ as the optimal capital stock of a firm with data. For any $A_{i,2}$, this solves:

$$K_2^d(A_{i,2}) = \arg\max_{K_2} \left[A_{i,2}(K_2)^{\alpha} - r_d K_2 \right] \tag{D.8}$$

We define $\bar{K}_2^{nd}$ and $\bar{K}_2^d$ as the cross-sectional expectations of the capital stocks chosen by firms without data and firms with data, respectively. Further, we define $\bar{Y}_2^{nd}$ and $\bar{Y}_2^d$ as the

cross-sectional expectations of the output levels of firms without data and firms with data, respectively. Lastly, we define Y_2 as the cross-sectional expectation of total output.

We begin by studying how a firm's predetermined access to data affects its responsiveness to aggregate shocks:

Proposition 8. *Consider the model with purely predictive data. If $\alpha < 0.5$, firms with data respond less strongly to aggregate productivity shocks, i.e., $\frac{\partial \bar{K}_2^{nd}/\partial \bar{A}}{\bar{K}_2^{nd}} > \frac{\partial \bar{K}_2^{d}/\partial \bar{A}}{\bar{K}_2^{d}}$ and $\frac{\partial \bar{Y}_2^{nd}/\partial \bar{A}}{\bar{Y}_2^{nd}} > \frac{\partial \bar{Y}_2^{d}/\partial \bar{A}}{\bar{Y}_2^{d}}$ holds.*

This result holds by the following logic: When firms have access to data about their idiosyncratic productivities, changes in aggregate productivity will induce smaller changes in their information sets, thereby eliciting a smaller response.

Endogenous access to data. The model can be extended to allow for endogenous data acquisition as follows: Suppose there is a finite number of markets, and each firm chooses how many markets to operate in. Entry costs are convex in the number of markets entered, implying that total profits are concave in the number of markets it enters. On each market in which it operates, a firm obtains access to data (as defined above) with an interior probability. Firms can pay a cost to increase this probability, i.e., to acquire superior data. Probabilistically, a firm that acquires better data therefore has access to superior information on a larger fraction of the markets in which it operates.

As in the main analysis, there is a strategic complementarity between a firm's size and its data acquisition choices: A firm that is active on more markets has stronger incentives to acquire data, given that the benefits of access to superior data accrue per market. Conversely, superior data raises the marginal return of expanding into additional markets. This complementarity implies that any aggregate shock will have a second-round effect on a firm's total investment that is qualitatively analogous to the second-round effect we highlighted in the main analysis. Thus, the possibility that firms can acquire data increases the responsiveness of any firm to an aggregate shock, and particularly so for firms that are already data-rich exogenously. Hence, the results we derived in the baseline analysis extend qualitatively.

D.4 Non-constant Returns to Scale in Data

In this section, we consider a model in which data may have increasing or decreasing returns to scale. Formally, this means that the effect of a given improvement in data access on a firm's productivity distribution depends on the firm's access to data. The presence of increasing returns to scale in data makes the relationship between a firm's predetermined

access to data and its responsiveness to an aggregate shock more positive. By contrast, the presence of decreasing returns to scale in data makes the relationship between a firm's access to data and its responsiveness to an aggregate shock more negative. This is because the presence of decreasing returns to scale in data reduces the magnitude of the second-round effect, and particularly so for data-rich firms. We present sufficient conditions under which our key results extend even when data has decreasing returns to scale—specifically, the results extend if the curvature of the map between data and the productivity distribution features relatively low curvature at high levels of data access.

Model. We consider a model that is analogous to our baseline model, with one key exception: We now specify that data affects the variance of a firm's productivity as follows:

$$VAR_t[A_{i,t+j}] = \bar{V}_{t+j} + \kappa_v \Delta(\sigma_{i,t+j}), \tag{D.9}$$

where $\Delta(\cdot)$ is an increasing function and $\sigma_{i,t+j} = \bar{\sigma}_i - \zeta_i K_{i,t+j}$. This model is equivalent to the model we considered in Section 4 if $\Delta(x) = x$. In contrast to the main analysis, we allow $\Delta''(\cdot)$ to be strictly positive or strictly negative. To fix ideas, we also assume that $\Delta'''(\cdot) = 0$.

For simplicity, we set $\zeta_i > 0$ (i.e., assume that the data feedback loop is active), $\eta_i = 0$ (i.e., abstract from the possibility that firms can directly acquire data), $\kappa_e = 0$ (i.e., assume that access to superior data only benefits firms by reducing their uncertainty), and $\delta = 1$. Everything else is as in the baseline model.

We say there are *increasing returns to scale in data* if $\Delta''(\cdot) < 0$ and *decreasing returns to scale in data* if $\Delta''(\cdot) > 0$. To see why this interpretation is appropriate, note that better data corresponds to lower $\bar{\sigma}_i$, which means that the curvature of $\Delta(\cdot)$ maps inversely into returns to scale in data, i.e:

$$\frac{\partial VAR_t[A_{i,t+1}]}{\partial \bar{\sigma}_i} = \kappa_v \Delta'(\bar{\sigma}_i - \zeta_i K_{i,t+1}) > 0 \quad ; \quad \frac{\partial^2 VAR_t[A_{i,t+1}]}{\partial \bar{\sigma}_i^2} = \kappa_v \Delta''(\bar{\sigma}_i - \zeta_i K_{i,t+1})$$

If $\Delta''(\cdot) > 0$, then the uncertainty reduction granted by superior access to data is particularly large (in magnitude) when $\bar{\sigma}_i$ is large, i.e., when a firm has limited access to data. This corresponds to decreasing returns to scale in data.

Analysis. If data has increasing returns to scale, we obtain the following result.

Proposition 9. *Suppose that data has increasing returns to scale. Then,* $\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i} < 0$ *holds.*

If data has increasing returns to scale, a given increase in the data acquired by a firm (the initial part of the second-round effect) leads to a relatively large improvement in the

productivity distribution of data-rich firms. Thus, the presence of increasing returns to scale in data increases the relative magnitude of the second-round effect for data-rich firms, which makes the relationship between firms' predetermined data access and their responsiveness to an aggregate shock more positive.

We now consider the case in which data has decreasing returns to scale:

Proposition 10. *Suppose that data has decreasing returns to scale. Then,* $\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i} < 0$ *holds if and only if:*

$$2\rho_i \kappa_v \zeta_i \Delta'(\bar{\sigma}_i - \zeta_i K^*_{t+1})(2-\alpha)(K^*_{t+1})^{1-\alpha} \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i}$$

$$+\rho_i \kappa_v \zeta_i \Delta''(\bar{\sigma}_i - \zeta_i K^*_{t+1})(K^*_{t+1})^{2-\alpha} \underbrace{\left[2\left(1 - \zeta_i \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i}\right) - (3-\alpha)\zeta_i \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} \right]}_{>0} < 0 \qquad \text{(D.10)}$$

To understand this result, note that $\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i} < 0$ holds if $\Delta'' = 0$ (and $\kappa_e = 0$), as in the main analysis. If data has decreasing returns to scale, the sign of $\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i}$ is no longer pinned down. This is because there is a new countervailing effect: Given that data has decreasing returns to scale, a given increase in the data that is acquired by a firm (the initial part of the second-round effect) leads to a relatively small improvement in the productivity distribution for data-rich firms. Thus, the presence of decreasing returns to scale in data dampens the relative magnitude of the second-order effect for data-rich firms. This effect thus may reverse the sign of the relationship between a firm's data access and its responsiveness to an aggregate productivity shock.

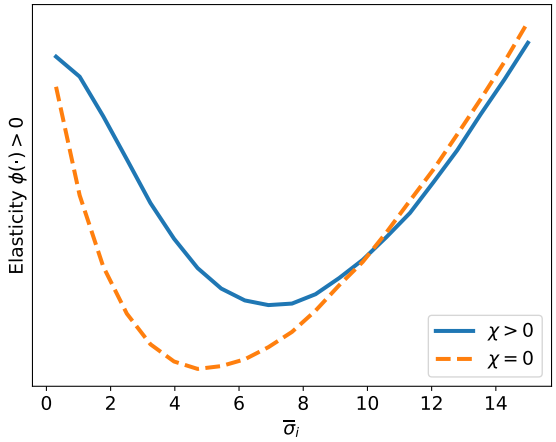
Under suitable parameter values, one can still match our empirical result that the relationship between firms' predetermined data access and their responsiveness to an aggregate productivity shock is negative within the subsample of data-poor firms and positive for data-rich firms. This result emerges (even if $\kappa_e = 0$) if the curvature of $\Delta(\cdot)$ is particularly large at high $\bar{\sigma}_i$ (i.e., for firms with limited access to data), whereas $\Delta(\cdot)$ is relatively linear if $\bar{\sigma}_i$ is small (i.e., for firms with access to high-quality data). Formally, one can show that:

$$\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i} > 0 \iff \frac{\Delta''(\bar{\sigma}_i - \zeta_i K^*_{t+1})}{\Delta'(\bar{\sigma}_i - \zeta_i K^*_{t+1})} > -\frac{2(2-\alpha)\frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i}}{(K^*_{t+1})\left[2\left(1 - \zeta_i \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i}\right) - (3-\alpha)\zeta_i \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i}\right]} \qquad \text{(D.11)}$$

To understand this condition, note that the right-hand side of the second inequality is positive if $\frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} < 0$. If $\Delta''(x) = 0$, $\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i} < 0$ must thus hold, i.e., data-rich firms respond more strongly to aggregate productivity shock (as in the main analysis, if $\kappa_e = 0$).[17]

[17] Note that $\frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} < 0$ holds in this case.

Figure D.3: Decreasing Returns to Data: Aggregate Productivity Shock Elasticity



This figure shows the elasticity $\phi(\cdot)$ for different data intensities $\bar{\sigma}_i$, in the model with decreasing returns to data (blue-solid line) and the model with constant returns to data (orange-dashed line). Calibration: $\beta = 0.96, \alpha = 0.5, r = 0.15, \bar{A} = 1, \bar{V} = 1, \zeta = 0.01, \kappa_e = 0.01, \kappa_v = 0.1, \rho = 0.5$ and $\chi = 0.1$ for the case with $\chi > 0$.

We complement the preceding analytical results using numerical simulations that are based on value function iteration, and where we relax the restriction $\kappa_e = 0$. The function $\Delta(x)$ is assumed to have the functional form $\Delta(x) = \chi x^2 + x$, where $x = \bar{\sigma} - \zeta_i K_{i,j+1}$, and $\chi \geq 0$. If $\chi = 0$, we are back to our baseline model. If $\chi > 0$, the model features decreasing returns to data. Figure D.3 shows the elasticity of the firm's optimal capital stock for different $\bar{\sigma}_i$ on the horizontal axes. The blue-solid line shows the case with decreasing returns to data, and the orange-dashed line the results for our baseline model. The qualitative patterns are analogous in either setup.

D.5 Proofs: Extensions

Proof of Proposition 7:

(i) Optimization problem and preliminaries.

As before, we consider a one-time (and transitory) aggregate productivity shock. We set $\bar{A}_{t+1} = \bar{A}$, and consider a shock to it. The Bellman equation reads:

$$\mathcal{V}(K_{i,t}) = \max_{K_{i,t+1}} \left\{ f(K_{i,t}, K_{i,t+1}) + \beta \mathcal{V}'(K_{i,t+1}) \right\},$$

where

$$f(K_{i,t}, K_{i,t+1}) = \left(\bar{A} - \kappa_e(\bar{\sigma}_i - \zeta_i K_{i,t+1})\right)(K_{i,t+1})^{\alpha} -$$

$$\big(\lambda_i \bar{V}_{t+1} + \lambda_i \kappa_v (\bar{\sigma}_i - \zeta_i K_{i,t+1})\big)\big(K_{i,t+1}\big)^{2\alpha} - r\big(K_{i,t+1} - (1-\delta)K_{i,t}\big) \tag{D.12}$$

and $\frac{\partial \mathcal{V}'(K_{i,t+1})}{\partial K_{i,t+1}} = r(1-\delta)$. The first-order condition thus reads:

$$\big(\bar{A} - \kappa_e(\bar{\sigma}_i - \zeta_i K_{i,t+1})\big)\alpha(K_{i,t+1})^{\alpha-1} + \big(\kappa_e \zeta_i\big)(K_{i,t+1})^{\alpha} - \big(-\lambda_i \kappa_v \zeta_i\big)\big(K_{i,t+1}\big)^{2\alpha}$$

$$-\big(\lambda_i \bar{V}_{t+1} + \lambda_i \kappa_v (\bar{\sigma}_i - \zeta_i K_{i,t+1})\big)(2\alpha)\big(K_{i,t+1}\big)^{2\alpha-1} - r + \beta\big[r(1-\delta)\big] = 0$$

Thus, the following condition has to hold in optimum:

$$T(\cdot) = \big(\bar{A} - \kappa_e \bar{\sigma}_i\big)\alpha(K^*_{t+1}(\cdot))^{\alpha-1} + (\alpha+1)\big(\kappa_e \zeta_i\big)(K^*_{t+1}(\cdot))^{\alpha} - r + \beta(1-\delta)r$$

$$-\big(\lambda_i \bar{V}_{t+1} + \lambda_i \kappa_v \bar{\sigma}_i\big)(2\alpha)\big(K^*_{t+1}(\cdot)\big)^{2\alpha-1} + \big(2\alpha+1\big)\big(\lambda_i \kappa_v \zeta_i\big)\big(K^*_{t+1}(\cdot)\big)^{2\alpha} = 0$$

This implies that:

$$\frac{\partial T}{\partial \bar{A}} = \alpha(K^*_{t+1}(\cdot))^{\alpha-1}$$

$$\frac{\partial T}{\partial K_{t+1}} = \big(\bar{A} - \kappa_e \bar{\sigma}_i\big)\alpha(\alpha-1)(K^*_{t+1}(\cdot))^{\alpha-2} + \alpha(\alpha+1)\big(\kappa_e \zeta_i\big)(K^*_{t+1}(\cdot))^{\alpha-1} +$$

$$-\big(\lambda_i \bar{V}_{t+1} + \lambda_i \kappa_v \bar{\sigma}_i\big)(2\alpha)(2\alpha-1)\big(K^*_{t+1}(\cdot)\big)^{2\alpha-2} + (2\alpha)\big(2\alpha+1\big)\big(\lambda_i \kappa_v \zeta_i\big)\big(K^*_{t+1}(\cdot)\big)^{2\alpha-1}$$

(ii) The relative effects of shocks when $\zeta_i = 0$.

Suppose $\alpha = 1/2$ and set $\zeta_i = 0$. Then, a closed form expression for $K^*_{t+1}(\cdot)$ is available, which solves:

$$\big(\bar{A} - \kappa_e \bar{\sigma}_i\big)\alpha(K^*_{t+1}(\cdot))^{\alpha-1} = \big(\lambda_i \bar{V}_{t+1} + \lambda_i \kappa_v \bar{\sigma}_i\big)(2\alpha)\big(K^*_{t+1}(\cdot)\big)^{2\alpha-1} + r - \beta(1-\delta)r \iff$$

$$K^*_{t+1}(\cdot) = \left[\frac{0.5\big(\bar{A} - \kappa_e \bar{\sigma}_i\big)}{r - \beta(1-\delta)r + \lambda_i \bar{V}_{t+1} + \lambda_i \kappa_v \bar{\sigma}_i}\right]^2 \tag{D.13}$$

Note that:

$$\frac{\partial K^*_{t+1}}{\partial \bar{A}} = \left[\frac{0.5\big(\bar{A} - \kappa_e \bar{\sigma}_i\big)}{r - \beta(1-\delta)r + \lambda_i \bar{V}_{t+1} + \lambda_i \kappa_v \bar{\sigma}_i}\right]^2 2\big(\bar{A} - \kappa_e \bar{\sigma}_i\big)^{-1} = \big(K^*_{t+1}(\cdot)\big)2\big(\bar{A} - \kappa_e \bar{\sigma}_i\big)^{-1}$$

Thus, the relative effect of an aggregate productivity shock is given by:

$$\phi(\cdot) = \frac{2\bar{A}}{\bar{A} - \kappa_e \bar{\sigma}_i} > 0$$

It follows that $\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i} \geq 0$, with a strict inequality if $\kappa_e > 0$.

(iii) The relative effects of an aggregate productivity shock when $\zeta_i > 0$ and $\kappa_e = 0$.

Under our assumptions, we have:

$$\frac{\partial K^*_{t+1}}{\partial \bar{A}} = \frac{-\alpha (K^*_{t+1}(\cdot))^{\alpha-1}}{\bar{A}\alpha(\alpha-1)(K^*_{t+1}(\cdot))^{\alpha-2} + (2\alpha)\big(2\alpha+1\big)\big(\lambda_i \kappa_v \zeta_i\big)\big(K^*_{t+1}(\cdot)\big)^{2\alpha-1}}$$

It follows that:

$$\phi(\cdot) = \frac{-\alpha \bar{A} (K^*_{t+1}(\cdot))^{\alpha-2}}{\bar{A}\alpha(\alpha-1)(K^*_{t+1}(\cdot))^{\alpha-2} + (2\alpha)\big(2\alpha+1\big)\big(\lambda_i \kappa_v \zeta_i\big)\big(K^*_{t+1}(\cdot)\big)^{2\alpha-1}}$$

$$\Longleftrightarrow$$

$$\phi(\cdot) = \frac{-\alpha \bar{A}}{\bar{A}\alpha(\alpha-1) + (2\alpha)\big(2\alpha+1\big)\big(\lambda_i \kappa_v \zeta_i\big)\big(K^*_{t+1}(\cdot)\big)^{\alpha+1}}$$

This implies that:

$$\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i} = \frac{\alpha \bar{A}}{\big[\bar{A}\alpha(\alpha-1) + (2\alpha)\big(2\alpha+1\big)\big(\lambda_i \kappa_v \zeta_i\big)\big(K^*_{t+1}(\cdot)\big)^{\alpha+1}\big]^2}$$

$$\left[(2\alpha)\big(2\alpha+1\big)(\alpha+1)\big(\lambda_i \kappa_v \zeta_i\big)\big(K^*_{t+1}(\cdot)\big)^{\alpha} \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i}\right] < 0$$

In the following, we show that $\frac{\partial^2 \phi(\cdot)}{\partial \bar{\sigma}_i^2} > 0$. To see why this holds true, note that:

$$\frac{\partial^2 \phi(\cdot)}{\partial \bar{\sigma}_i^2} = \frac{-2\alpha \bar{A}}{\big[\bar{A}\alpha(\alpha-1) + (2\alpha)\big(2\alpha+1\big)\big(\lambda_i \kappa_v \zeta_i\big)\big(K^*_{t+1}(\cdot)\big)^{\alpha+1}\big]^3}$$

$$\left[(2\alpha)\big(2\alpha+1\big)(\alpha+1)\big(\lambda_i \kappa_v \zeta_i\big)\big(K^*_{t+1}(\cdot)\big)^{\alpha} \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i}\right]^2 +$$

$$\frac{\alpha \bar{A}}{\big[\bar{A}\alpha(\alpha-1) + (2\alpha)\big(2\alpha+1\big)\big(\lambda_i \kappa_v \zeta_i\big)\big(K^*_{t+1}(\cdot)\big)^{\alpha+1}\big]^2}$$

$$\left[(2\alpha)\big(2\alpha+1\big)(\alpha+1)\alpha\big(\lambda_i \kappa_v \zeta_i\big)\big(K^*_{t+1}(\cdot)\big)^{\alpha-1}\left(\frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i}\right)^2 + (2\alpha)\big(2\alpha+1\big)(\alpha+1)\big(\lambda_i \kappa_v \zeta_i\big)\big(K^*_{t+1}(\cdot)\big)^{\alpha} \frac{\partial^2 K^*_{t+1}}{\partial \bar{\sigma}_i^2}\right]$$

Note that $\frac{\partial T}{\partial K_{t+1}}(K_{t+1})^{2-\alpha}$ is strictly negative under our assumptions. Thus, the expression for $\frac{\partial^2 \phi(\cdot)}{\partial \bar{\sigma}_i^2}$ is strictly positive if $\frac{\partial^2 K^*_{t+1}}{\partial \bar{\sigma}_i^2} > 0$. To see that this holds true, note that:

$$\frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} = -\frac{-\lambda_i \kappa_v}{-0.25\bar{A}(K^*_{t+1}(\cdot))^{-1.5} + 2\lambda_i \kappa_v \zeta_i} = \frac{\lambda_i \kappa_v}{-0.25\bar{A}(K^*_{t+1}(\cdot))^{-1.5} + 2\lambda_i \kappa_v \zeta_i} < 0$$

Furthermore, we have that:

$$\frac{\partial^2 K^*_{t+1}}{\partial \bar{\sigma}_i^2} = \frac{-\lambda_i \kappa_v}{\left[-0.25\bar{A}(K^*_{t+1}(\cdot))^{-1.5} + 2\lambda_i \kappa_v \zeta_i\right]^2} \left[\frac{3}{8}\bar{A}(K^*_{t+1})^{-2.5}\frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i}\right] > 0$$

■

Proof of Proposition 8:

Optimal capital choices and cross-sectional means. A firm with data observes $A_{i,2}$ when choosing $K_{i,2}$ and solves

$$\max_{K_2} \; A_{i,2}K_2^\alpha - r_d K_2, \qquad r_d := r + \tilde{\rho}_d.$$

The first-order condition yields

$$\alpha A_{i,2} K_2^{\alpha-1} = r_d,$$

which yields $K_2^d(A_{i,2}) = \left(\frac{\alpha A_{i,2}}{r_d}\right)^{\frac{1}{1-\alpha}}$. Taking expectations across firms gives

$$\bar{K}_2^d = \alpha^{\frac{1}{1-\alpha}} \, r_d^{\frac{1}{\alpha-1}} \, \mathbb{E}\left[A_{i,2}^{\frac{1}{1-\alpha}}\right]. \tag{D.14}$$

The output of a firm with data is $Y_{i,2}^d = A_{i,2}(K_2^d(A_{i,2}))^\alpha$. This implies that

$$\bar{Y}_2^d = \alpha^{\frac{\alpha}{1-\alpha}} \, r_d^{\frac{\alpha}{\alpha-1}} \, \mathbb{E}\left[A_{i,2}^{\frac{1}{1-\alpha}}\right]. \tag{D.15}$$

A firm without data chooses capital under uncertainty and solves

$$\max_{K_2} \; \mathbb{E}[A_{i,2}]K_2^\alpha - r_{nd} K_2, \qquad r_{nd} := r + \tilde{\rho}_{nd}.$$

This yields $K_2^{nd} = \left(\frac{\alpha \mathbb{E}[A_{i,2}]}{r_{nd}}\right)^{\frac{1}{1-\alpha}}$. It follows that

$$\bar{Y}_2^{nd} = \mathbb{E}[A_{i,2}](K_2^{nd})^{\alpha} = \alpha^{\frac{\alpha}{1-\alpha}} r_{nd}^{\frac{\alpha}{\alpha-1}} \left(\mathbb{E}[A_{i,2}]\right)^{\frac{1}{1-\alpha}}. \tag{D.16}$$

Aggregate productivity shocks. Note that $A_{i,2} = \bar{A} + \epsilon_i$. For firms without data,

$$\frac{\partial \bar{Y}_2^{nd}/\partial \bar{A}}{\bar{Y}_2^{nd}} = \frac{1}{1-\alpha}\frac{1}{\mathbb{E}[A_{i,2}]}.$$

For firms with data,

$$\frac{\partial \bar{Y}_2^{d}/\partial \bar{A}}{\bar{Y}_2^{d}} = \frac{1}{1-\alpha}\frac{\mathbb{E}\left[A_{i,2}^{\frac{\alpha}{1-\alpha}}\right]}{\mathbb{E}\left[A_{i,2}^{\frac{1}{1-\alpha}}\right]}.$$

Let $q := \frac{\alpha}{1-\alpha}$. If $\alpha < 1/2$, then $q \in (0,1)$. Since $x \mapsto x^q$ is strictly concave and $x \mapsto x^{1+q}$ is strictly convex on $\mathbb{R}_+$, Jensen's inequality implies

$$\mathbb{E}[A^q] < (\mathbb{E}[A])^q, \qquad \mathbb{E}[A^{1+q}] > (\mathbb{E}[A])^{1+q}.$$

Combining these yields

$$\frac{\mathbb{E}[A^q]}{\mathbb{E}[A^{1+q}]} < \frac{(\mathbb{E}[A])^q}{(\mathbb{E}[A])^{1+q}} = \frac{1}{\mathbb{E}[A]},$$

which implies

$$\frac{\partial \bar{Y}_2^{nd}/\partial \bar{A}}{\bar{Y}_2^{nd}} > \frac{\partial \bar{Y}_2^{d}/\partial \bar{A}}{\bar{Y}_2^{d}}.$$

The same holds for capital. This proves the proposition. ■

Proof of Proposition 9: The Bellman equation reads:

$$V(K_t) = \max_{K_{t+1}} \left[f_t(K_{t+1}) + \beta V'(K_{t+1}) \right], \tag{D.17}$$

where

$$f_t(K_{i,t+1}) := \bar{A}(K_{i,t+1})^{\alpha} - \left(r + \rho_i \bar{V} + \rho_i \kappa_v \Delta(\bar{\sigma}_i - \zeta_i K_{i,t+1})\right) K_{i,t+1}$$

Given that $\delta = 1$, $\frac{\partial V'(K_{t+1})}{\partial K_{t+1}} = 0$ holds. Thus, the first-order condition reads:

$$T(\cdot) = \bar{A}\alpha(K_{t+1}^*)^{\alpha-1} - \left(r + \rho_i \bar{V} + \rho_i \kappa_v \Delta(\bar{\sigma}_i - \zeta_i K_{t+1}^*)\right) + \left(\rho_i \kappa_v \zeta_i \Delta'(\bar{\sigma}_i - \zeta_i K_{t+1}^*)\right) K_{t+1}^* = 0$$

Thus, we have:

$$\frac{\partial T}{\partial \bar{\sigma}_i} = -\rho_i \kappa_v \Delta'(\bar{\sigma}_i - \zeta_i K^*_{t+1}) + \rho_i \kappa_v \zeta_i \Delta''(\bar{\sigma}_i - \zeta_i K^*_{t+1}) K^*_{t+1}$$

$$\frac{\partial T}{\partial \bar{A}} = \alpha (K^*_{t+1})^{\alpha-1}$$

$$\frac{\partial T}{\partial K_{t+1}} = \bar{A}\alpha(\alpha-1)(K^*_{t+1})^{\alpha-2} + 2\rho_i \kappa_v \zeta_i \Delta'(\bar{\sigma}_i - \zeta_i K^*_{t+1}) - \rho_i \kappa_v (\zeta_i)^2 \Delta''(\bar{\sigma}_i - \zeta_i K^*_{t+1}) K^*_{t+1}$$

Assumption 1 implies that $\frac{\partial T}{\partial K_{t+1}} < 0$. Note further that $\frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} < 0$ holds if data has increasing returns to scale.

Aggregate productivity shocks:

The relative effect of an aggregate productivity shock is thus:

$$\phi(\cdot) = -\frac{\alpha (K^*_{t+1})^{\alpha-1} \bar{A} (K^*_{t+1})^{-1}}{\bar{A}\alpha(\alpha-1)(K^*_{t+1})^{\alpha-2} + 2\rho_i \kappa_v \zeta_i \Delta'(\bar{\sigma}_i - \zeta_i K^*_{t+1}) - \rho_i \kappa_v (\zeta_i)^2 \Delta''(\bar{\sigma}_i - \zeta_i K^*_{t+1}) K^*_{t+1}} =$$

$$-\frac{\alpha \bar{A}}{\bar{A}\alpha(\alpha-1) + 2\rho_i \kappa_v \zeta_i \Delta'(\bar{\sigma}_i - \zeta_i K^*_{t+1})(K^*_{t+1})^{2-\alpha} - \rho_i \kappa_v (\zeta_i)^2 \Delta''(\bar{\sigma}_i - \zeta_i K^*_{t+1})(K^*_{t+1})^{3-\alpha}}$$

Note that $\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i} > 0$ holds if and only if $\frac{\partial \phi_d(\cdot)}{\partial \bar{\sigma}_i} > 0$, where:

$$\phi_d(\cdot) := \bar{A}\alpha(\alpha-1) + 2\rho_i \kappa_v \zeta_i \Delta'(\bar{\sigma}_i - \zeta_i K^*_{t+1})(K^*_{t+1})^{2-\alpha} - \rho_i \kappa_v (\zeta_i)^2 \Delta''(\bar{\sigma}_i - \zeta_i K^*_{t+1})(K^*_{t+1})^{3-\alpha}$$

We have:

$$\frac{\partial \phi_d(\cdot)}{\partial \bar{\sigma}_i} =$$

$$2\rho_i \kappa_v \zeta_i \Delta'(\bar{\sigma}_i - \zeta_i K^*_{t+1})(2-\alpha)(K^*_{t+1})^{1-\alpha} \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} + 2\rho_i \kappa_v \zeta_i \Delta''(\bar{\sigma}_i - \zeta_i K^*_{t+1}) \left(1 - \zeta_i \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i}\right)(K^*_{t+1})^{2-\alpha}$$

$$-(3-\alpha)\rho_i \kappa_v (\zeta_i)^2 \Delta''(\bar{\sigma}_i - \zeta_i K^*_{t+1})(K^*_{t+1})^{2-\alpha} \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} - \rho_i \kappa_v (\zeta_i)^2 \Delta'''(\bar{\sigma}_i - \zeta_i K^*_{t+1}) \left(1 - \zeta_i \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i}\right)(K^*_{t+1})^{3-\alpha}$$

$$\Longleftrightarrow$$

$$\frac{\partial \phi_d(\cdot)}{\partial \bar{\sigma}_i} = 2\rho_i \kappa_v \zeta_i \Delta'(\bar{\sigma}_i - \zeta_i K^*_{t+1})(2-\alpha)(K^*_{t+1})^{1-\alpha} \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i}$$

$$+\rho_i \kappa_v \zeta_i \Delta''(\bar{\sigma}_i - \zeta_i K^*_{t+1})(K^*_{t+1})^{2-\alpha} \underbrace{\left[2\left(1 - \zeta_i \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i}\right) - (3-\alpha)\zeta_i \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} \right]}_{>0}$$

If data has increasing returns to scale, then $\Delta''(x) < 0$ and $\frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} < 0$ hold. Thus, all terms are strictly negative. This implies that $\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i} < 0$

■

Proof of Proposition 10: Note that $\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i} > 0$ holds if and only if $\frac{\partial \phi_d(\cdot)}{\partial \bar{\sigma}_i} > 0$. By the arguments made in the previous proposition, $\frac{\partial \phi(\cdot)}{\partial \bar{\sigma}_i} < 0$ thus holds if and only if:

$$\frac{\partial \phi_d(\cdot)}{\partial \bar{\sigma}_i} = 2\rho_i \kappa_v \zeta_i \Delta'(\bar{\sigma}_i - \zeta_i K^*_{t+1})(2-\alpha)(K^*_{t+1})^{1-\alpha} \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i}$$

$$+\rho_i \kappa_v \zeta_i \Delta''(\bar{\sigma}_i - \zeta_i K^*_{t+1})(K^*_{t+1})^{2-\alpha} \left[2\left(1 - \zeta_i \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i}\right) - (3-\alpha)\zeta_i \frac{\partial K^*_{t+1}}{\partial \bar{\sigma}_i} \right] < 0$$

■

E General Equilibrium Model

This appendix details the general equilibrium model of Section 5. We embed the data block of Section 3 into a New Keynesian wholesale–retail economy with a representative data-using producer and Rotemberg price setters, in the spirit of Winberry et al. (2025). Throughout, D_t denotes the stock of data, related to the noise object of the main analysis by

$$\sigma_t = \bar{\sigma} - D_t. \tag{E.1}$$

The local linearization of the analytical relationships, $\mathbb{E}_t[A_{t+1}] = \bar{A}_{t+1} + \kappa_e D_{t+1}$ and $VAR_t[A_{t+1}] = \bar{V}_{t+1} - \kappa_v D_{t+1}$, motivates the globally positive specifications used below.

E.1 The data economy in general equilibrium

Data-augmented total factor productivity and the data-dependent investment cost wedge are given by

$$A_t = \exp\big(a_t + \kappa_e D_t\big), \tag{E.2}$$

$$\Omega_t = \omega \exp\big(v_t - \kappa_v D_{t+1}\big), \qquad \Omega_D(\cdot) < 0, \tag{E.3}$$

where a_t is an aggregate productivity shifter and v_t an aggregate uncertainty shifter. As in the main text, $\kappa_e \geq 0$ governs the effect of data on expected productivity and $\kappa_v \geq 0$ its

effect on the conditional variance, which here maps into the cost of installing capital. Under the exponential specification (E.3), we have

$$\Omega_D(D_{t+1}, v_t) = -\kappa_v \, \Omega_t. \tag{E.4}$$

Data accumulates through persistence, direct acquisition, and the feedback loop:

$$D_{t+1} = (1 - \delta_D) \, D_t + \eta \, d_t + \zeta \, K_{t+1}, \tag{E.5}$$

with d_t denoting directly acquired data, $\delta_D \in [0, 1]$ the data depreciation rate, $\eta \geq 0$ the efficiency of direct acquisition, and $\zeta \geq 0$ the strength of the data feedback loop.

E.2 Households, final good, and policy

The representative household solves[18]

$$\max_{\{C_t, N_t, B_t\}} \; \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\tilde{\sigma}}}{1-\tilde{\sigma}} - \tilde{\chi} \frac{N_t^{1+\varphi}}{1+\varphi} \right] \tag{E.6}$$

subject to the real budget constraint

$$C_t + \frac{B_t}{P_t} = w_t N_t + \frac{R_{t-1}}{\Pi_t} \frac{B_{t-1}}{P_{t-1}} + \Pi_t^{agg} - T_t, \tag{E.7}$$

where $w_t \equiv W_t / P_t$, $\Pi_t \equiv P_t / P_{t-1}$, and Π_t^{agg} are aggregate real profits. The optimality conditions are

$$1 = \mathbb{E}_t \left[\Lambda_{t,t+1} \frac{R_t}{\Pi_{t+1}} \right], \qquad \Lambda_{t,t+1} \equiv \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\tilde{\sigma}}, \tag{E.8}$$

$$\tilde{\chi} N_t^{\varphi} = w_t C_t^{-\tilde{\sigma}}. \tag{E.9}$$

A competitive final-good producer aggregates retail varieties,

$$Y_t = \left(\int_0^1 y_t(j)^{\frac{\varepsilon-1}{\varepsilon}} \, dj \right)^{\frac{\varepsilon}{\varepsilon-1}}, \quad P_t = \left(\int_0^1 P_t(j)^{1-\varepsilon} \, dj \right)^{\frac{1}{1-\varepsilon}}, \quad y_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\varepsilon} Y_t. \tag{E.10}$$

[18] For the case $\tilde{\sigma} = 1$, consumption utility is given by $log(C_t)$.

Monetary policy follows a Taylor rule with interest rate smoothing

$$\log\left(\frac{R_t}{\bar{R}}\right) = \rho_R \log\left(\frac{R_{t-1}}{\bar{R}}\right) + (1-\rho_R)\,\varphi_\pi \log\left(\frac{\Pi_t}{\bar{\Pi}}\right). \tag{E.11}$$

E.3 Sticky-price retailers

Retailer j buys the homogeneous wholesale good at real relative price $s_t \equiv P_t^w/P_t$ and sets $P_t(j)$ subject to a Rotemberg cost. Its real profit is

$$\Pi_t^R(j) = \left(\frac{P_t(j)}{P_t} - s_t\right) y_t(j) - \frac{\varphi_p}{2}\left(\frac{P_t(j)}{\bar{\Pi}\,P_{t-1}(j)} - 1\right)^2 Y_t. \tag{E.12}$$

In a symmetric equilibrium the pricing condition yields the nonlinear New Keynesian Phillips curve

$$1 - \varepsilon + \varepsilon s_t - \varphi_p\left(\frac{\Pi_t}{\bar{\Pi}} - 1\right)\frac{\Pi_t}{\bar{\Pi}} + \mathbb{E}_t\left[\Lambda_{t,t+1}\,\varphi_p \frac{Y_{t+1}}{Y_t}\left(\frac{\Pi_{t+1}}{\bar{\Pi}} - 1\right)\frac{\Pi_{t+1}}{\bar{\Pi}}\right] = 0, \tag{E.13}$$

with steady-state markup pricing $s = (\varepsilon - 1)/\varepsilon$.

E.4 The representative data-using producer

The representative producer's production function is

$$Y_t = A_t K_t^{\alpha} N_t^{1-\alpha}, \tag{E.14}$$

with A_t given by (E.2). The producer sells the homogeneous good at relative price s_t and produces with capital, labor, and data-augmented productivity (E.2). Capital evolves as

$$K_{t+1} = (1-\delta_K)K_t + I_t. \tag{E.15}$$

Direct data acquisition and capital adjustment are costly,

$$G(d_t) = \frac{\varphi_d}{2} d_t^2, \qquad \mathcal{A}(I_t, K_t) = \frac{\varphi_I}{2}\left(\frac{I_t}{K_t} - \delta_K\right)^2 K_t. \tag{E.16}$$

The producer chooses $\{N_t, I_t, K_{t+1}, d_t, D_{t+1}\}_{t\geq 0}$ to maximize discounted profits (using the household's SDF Λ)

$$\max\ \mathbb{E}_0 \sum_{t=0}^{\infty} \Lambda_{0,t}\Big[s_t Y_t - w_t N_t - I_t - \mathcal{A}(I_t, K_t) - G(d_t) - \Omega_t I_t\Big], \tag{E.17}$$

subject to (E.5), (E.2), and (E.15). Let q_t be the multiplier on capital accumulation and μ_t the multiplier on data accumulation. The optimality conditions are

$$w_t = s_t(1-\alpha)\frac{Y_t}{N_t}, \tag{E.18}$$

$$q_t = 1 + \Omega_t + \mathcal{A}_I(I_t, K_t), \qquad \mathcal{A}_I(I_t, K_t) = \varphi_I\left(\frac{I_t}{K_t} - \delta_K\right), \tag{E.19}$$

$$\varphi_d d_t = \eta\,\mu_t, \tag{E.20}$$

$$q_t = \zeta\,\mu_t + \mathbb{E}_t\left[\Lambda_{t,t+1}\left(s_{t+1}\alpha\frac{Y_{t+1}}{K_{t+1}} + (1-\delta_K)q_{t+1} - \mathcal{A}_K(I_{t+1}, K_{t+1})\right)\right], \tag{E.21}$$

$$\mu_t = -\Omega_D(D_{t+1}, v_t)I_t + \mathbb{E}_t[\Lambda_{t,t+1}\left(s_{t+1}\kappa_e Y_{t+1} + (1-\delta_D)\mu_{t+1}\right)], \tag{E.22}$$

where, for the quadratic adjustment cost in (E.16),

$$\mathcal{A}_K(I_t, K_t) \equiv \frac{\partial \mathcal{A}(I_t, K_t)}{\partial K_t} = \frac{\varphi_I}{2}\left[\delta_K^2 - \left(\frac{I_t}{K_t}\right)^2\right]. \tag{E.23}$$

Using (E.4), the data Euler equation (E.22) can be written compactly as

$$\mu_t = \underbrace{\kappa_v \Omega_t I_t}_{\text{data lowers current wedge}} + \mathbb{E}_t\left[\Lambda_{t,t+1}\left(\underbrace{s_{t+1}\kappa_e Y_{t+1}}_{\text{higher future TFP}} + (1-\delta_D)\mu_{t+1}\right)\right]. \tag{E.24}$$

Equations (E.21) and (E.24) are the general-equilibrium counterpart of the strategic complementarity in Lemma 1: capital raises the value of data through $\zeta\mu_t$, and data raises the value of capital by lowering the installation wedge through $\kappa_v\Omega_t I_t$ and by raising future productivity through $s_{t+1}\kappa_e Y_{t+1}$.

E.5 Equilibrium

An equilibrium is a sequence $\{C_t, N_t, Y_t, K_t, I_t, D_t, d_t, \Pi_t, s_t, w_t, q_t, \mu_t, R_t\}_{t\geq 0}$ satisfying household optimality (E.8)–(E.9), retail pricing (E.13), producer optimality (E.14), (E.5), (E.15),

(E.18)–(E.24), the policy rule (E.11), and market clearing. Goods-market clearing is

$$Y_t = C_t + I_t + \mathcal{A}(I_t, K_t) + G(d_t) + \Omega_t I_t + \frac{\varphi_p}{2}\left(\frac{\Pi_t}{\bar{\Pi}} - 1\right)^2 Y_t, \tag{E.25}$$

where the last two terms are real resource losses from the financing/installation friction and from price adjustment. Bonds are in zero net supply, $B_t = 0$, and aggregate profits paid to households are

$$\Pi_t^{agg} = Y_t - w_t N_t - I_t - \mathcal{A}(I_t, K_t) - G(d_t) - \Omega_t I_t - \frac{\varphi_p}{2}\left(\frac{\Pi_t}{\bar{\Pi}} - 1\right)^2 Y_t. \tag{E.26}$$

The exogenous driving processes are

$$a_t = \rho_a a_{t-1} + \varepsilon_t^a, \qquad v_t = \rho_v v_{t-1} + \varepsilon_t^v, \tag{E.27}$$

the general-equilibrium counterparts of the aggregate productivity and uncertainty shocks studied in the partial-equilibrium analysis.